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20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

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1 Abstract

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2 List of Abbreviations

AG	Canton of Aargau
ALG	Alternative – the Greens Zug (Alternative – die Grünen Zug)
BDP	Conservative Democratic Party
BGE	Decisions of the Swiss Federal Supreme Court (official collection)
CSV	Comma-separated values
CV	Coefficient of variation
CVP	Christian Democratic People's Party
DP	Double Pukelsheim system
EQ1	Exploratory question 1 (district magnitude)
FDP	FDP.The Liberals
FSC	Swiss Federal Supreme Court
GLP	Green Liberal Party
GR	Canton of Graubünden
HB	Hagenbach-Bischoff method
JFSH	Young Liberals Schaffhausen (Jungfreisinnige Schaffhausen)
JUSO	Young Socialists Switzerland
NW	Canton of Nidwalden
PDF	Portable Document Format
pp	percentage points
PR	proportional representation
Q1	First quartile (25th percentile)
Q3	Third quartile (75th percentile)
SH	Canton of Schaffhausen
SP	Social Democratic Party of Switzerland
SVP	Swiss People's Party
SZ	Canton of Schwyz
UR	Canton of Uri
VS	Canton of Valais
XLSX	Excel workbook file format (Office Open XML)
ZG	Canton of Zug
ZH	Canton of Zurich

3 Introduction

On the 18 December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 I 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat varied heavily from district to district. Further did some districts have thresholds for parties to win a seat of up to 34%. Consequently, votes did not have equal effect, which undermines the electoral equality included in Art. 8 (Equality before the law) and Art. 34 (political rights) of the Federal Constitution. Based on this criticism the court decided that the City of Zurich had to implement corrective measures, but could choose their form (BGE 129 I 185 E. 7.3, E. 7.6.2–7.6.3 and E. 8.3.2; Schweizerische Eidgenossenschaft, 1999, Arts. 8 and 34). The Government of the Canton of Zurich decided to keep the constituency structure and instead commissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a method called "biproportional divisor method with standard rounding" which is today often simply called "Double Pukelsheimer" or "Biproportional apportionment". (Regierungsrat des Kantons Zürich, 2014, pp. 3–4; Lüscher, 2024, pp. 9-10)

The distinguishing feature of the Pukelsheimer Proporz to other proportional representation systems, is the so called upper-apportionment. It artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between seat and voteshares gets minimized and with that also the effective threshold for parties to enter the parliament. (Lüscher, 2023, p. 575)

Through its upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties, since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz's Landrat election in 2016, there under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they received three seats (Lüscher, 2025, p. 1-3).

Since Zurich first used the system in 2007, eight further cantons have used it: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022). In 2025 also the voters from Basel Landschaft decided to implement the biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While nine of these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. In Thurgau, the initiative "Mehr Transparenz bei Wahlen im Thurgau" is currently pending at the Grand Council of the Canton and a special parliamentary committee is examining the iniatitive, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kantons Thurgau, n.d.; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to adapt the Pukelsheimer algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, suggests that the decision by the FSC had a substantial impact on how Swiss citizens elect their regional parliaments today. It seems that the double-Pukelsheimer through

its wide implementation across cantons is a success story. Since the first election under double Pukelsheim in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. How much has the reform changed smaller parties' seat shares and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim system? Are there other major differences, for example in how the reform performs in smaller versus larger parliaments, and how do smaller parties' seat shares and gains evolve across successive elections?

Existing research primarily examines Double Pukelsheim in legal, normative, or system-level terms. A few studies also conduct counterfactual comparisons, but these cover either individual elections or only one election per canton (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592; Leuzinger, 2018, p. 42; Lüscher, 2024, p. 13). Although their results indicate that smaller parties can benefit from Double Pukelsheim, the magnitude and consistency of this mechanical allocation difference across successive elections remain less systematically examined.

This thesis addresses this gap by applying a consistent Hagenbach–Bischoff counterfactual to 31 elections in eight transition cantons. Two Graubünden elections are presented as contextual comparisons but excluded from the main transition estimate. Holding the observed post-reform votes and list structure constant, the analysis measures the difference in the share of seats won by smaller electoral lists and examines variation across institutional settings and successive elections.

4 Theory and Literature Review

4.1 Institutional and Theoretical Background

This chapter provides the institutional and theoretical background for analyzing the transition from Hagenbach-Bischoff to Double Pukelsheim and reviews the relevant literature. It first outlines the historical development of electoral systems in Switzerland and then explains the functioning and limitations of the Hagenbach-Bischoff and Double Pukelsheim methods. It subsequently examines the adoption and canton-specific implementation of Double Pukelsheim. Finally, the literature review situates this thesis within previous research before the expected effects of the transition are derived.

4.1.1 Historical Development

Until the late nineteenth century swiss voters elected their regional parliaments only under majority voting. In a majority voting system, the candidates or parties who receive the relative or absolute majority receive the mandates, which leads to a systematic favourment of the

[1]The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c, 2026); Schwyz (2016, 2020, and 2024; Staatskanzlei Kanton Schwyz, 2016, 2020, 2024); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

dominant parties. Over the time, this inequality, led to dissatisfaction among smaller parties. Subsequently, these demandend, to change to a proportional electoral system. In 1891 the canton of Ticino, was the first one to use such a PR-System to vote for their Grand Council, during the following 30 years multiple cantonsfollowed. When the socialist and conservative minorities on their third attempt in 1918 achieved, that the National Council will be elected through a proportional system, it boosted the general adoption of the system across Switzerland. Most of the cantons Grand Councils aswell as the National Council were then elected with the Hagenbach-Bischoff or another proportional system. In the beginning of the 21th century, the cantons of Graubünden and Appenzell Innerrhoden were the only ones to still vote their parliaments with the majority system. (Gilg, 2014, "Mehrheits- und Verhältniswahl" section ; Bundeskanzlei, 2013, p. 2) Nowadays the Kanton Appenzell Innerrhoden is the only remaining canton voting the Grand Council by majority voting (Bundeskanzlei, 2025, p. 18). It can therefore be said, that proportional systems have become the dominating electoral systems for cantonal parliaments in Switzerland.

In the last 20 years the prior dominance of the Hagenbach-Bischoff system has been demolished by the wide implementation of the Pukelsheimer Algorithm. As of 2025, eleven cantons distribute the parliamentary seats with Hagenbach-Bischoff and ten cantons with the DP, two cantons namely Glarus and Basel-Landschaft use the Sainte-Laguë method and two namely Ticino and Vaud use the Hare/Niemeyer method (Bundeskanzlei, 2025, p.18). Both of the later named methods, are generally associated with higher proportionality than the HB system, therefore this might partly epxlain why none of the cantons using these two PR-Systems have adopted the Double Pukelsheim system (Bundeskanzlei, 2025, pp. 22; Deutscher Bundestag, 2007, p.9).This means that of the nine cantons analyzed here, eight previously used Hagenbach-Bischoof for at least the majority of their parliamentary seats and only Graubünden had used a different system with majority voting. With respect to, that Zug's Neuheim constituency and Uri's smaller municipalities also used Majorz rather than Hagenbach-Bischoff (Bundeskanzlei, 2013, Table 1; BGE 136 I 376 E. 5.1 ; Standeskanzlei Uri, 2020, p. 1; Statistisches Amt des Kantons Zürich, 2011, p. 286; Regierungsrat des Kantons Schaffhausen, 2007, p. 6; Regierungsrat des Kantons Aargau, 2007, p. 18).

4.1.2 The Hagenbach-Bischoff Method

To be able to comprehend later the differences between the HB and the DP Method one has to understand the allocation under the priorly used Hagenbach-Bischoff system.

Under Hagenbach-Bischoff, seats are allocated separately in each constituency. Let M denote the number of mandates in the constituency, v_i the votes for an eligible list i, and V the total list votes used to calculate the distribution number. Whereas the distribution number D is the smallest integer strictly greater than $V/(M+1)$. In the first allocation, each list receives the integer part of its votes divided by D (Bundeskanzlei, 2025, p. 41):

$$D = \left\lfloor \frac{V}{M+1} \right\rfloor + 1, \quad s_i = \left\lfloor \frac{v_i}{D} \right\rfloor.$$

If the first allocation leaves seats unassigned, the remaining seats are awarded one at a time. For each list, its votes are divided by the number of seats it currently holds plus one. The list with the highest quotient receives the next seat; its quotient h_i and the allocated seats s_i are then recalculated, and the process continues until all M seats have been filled:

$$h_i = \frac{v_i}{s_i + 1}.$$

It should be noted, that the HB system calculation produces the same allocation as the D'Hondt divisor method with downward rounding (Bundeskanzlei, 2025, p. 41). Further there are some factors, like list alliances, eligibility thresholds, and rules for resolving equal quotients which depend on the electoral law of the corresponding canton. The counterfactual in this thesis applies the relevant former cantonal rules within each constituency, as detailed in the Data Cleaning section.

The central weakness of the traditional cantonal application of Hagenbach–Bischoff does not arise solely from the allocation formula, but from applying it separately in constituencies that often have few seats. Small district magnitudes create high and across districts varying natural thresholds, meaning that votes for smaller parties may fail to contribute to a seat.This might hinder equality of electoral success and systematically disadvantage smaller parties. As shown in the later section on the adoption of Double Pukelsheim, these inequalities were central to the legal and political debates that led several cantons to adopt Double Pukelsheim. (Pukelsheim & Schuhmacher, 2004, pp. 507-508, 521; BGE 129 I 185 E. 7.1.1 and E. 7.6.2)

4.1.3 The Double Pukelsheimer Method

Based on the weaknesses of the Hagenbach-Bischoff Method and the FSC decision, the canton of Zurich comissioned the mathematician Friedrich Pukelsheim to develop a new method (Lüscher, 2024, p.10). The following section explores how the algorithm works and how it makes sure that it complies with the Federal Law and mitigates the HB weaknesses.

Based on Article 8 and 34 of the Federal Constitution, there are three dimensions of electoral equality, which have to be followed. These are principle of equal voting value, equality of voting power and equality of electoral value. The equal voting value ensures that each voter within an electoral district gets allocated the identical number of votes. The second principle is the equality of voting power (voting weight guarantee). This requires the population represented per seat to be approximately equal across electoral districts. Thirdly there is the equality of electoral success, which should assure, that all votes contribute equally to the election and that as many votes as possible are taken into account in the allocation of seats. (Pukelsheim & Schuhmacher, 2004, pp. 506-507) Districts with fewer than nine seats have a natural quorum above 10% under the approximation $1/(M+1)$, where M is the number of seats. The Federal Supreme Court treats 10% as a target for natural quorums when redesigning an electoral system. Exceptionally, compelling historical or regional minority-protection reasons may justify a higher natural quorum. (Pukelsheim & Schuhmacher, 2004, p. 505; BGE 131 I 74 E. 5.2–5.4) Equality of counting value concerns the formal treatment of votes, while equality of voting power depends on the separately determined ratio of population to seats in each electoral district (Pukelsheim & Schuhmacher, 2004, pp. 507, 515–516). For the traditional application of Hagenbach–Bischoff, the main difficulty lies in equality of electoral success when small districts impose high natural thresholds (Pukelsheim & Schuhmacher, 2004, p. 508; BGE 129 I 185 E. 7.1.1 and E. 7.6.2).

To approach this main issue of the Hagenbach-Bischoff, the Double Pukelsheimer or originally called the "Neues Zürcher Zuteilungsverfahren" knows a upper and a lower apportionment (Pukelsheim & Schuhmacher, 2004, p.513).

- **Upper Apportionment** To bypass the natural quorums, in the upper apportionment, the whole canton is seen as one single district. All the seats are then directly proportionally to the vote share allocated towards the parties. For that the votes from the district-level have to be summarized on a cantonal level. This leads to the first difficulty, which arises through the variation in district magnitude and the resulting circumstance that voters receive across constituencies a different amount of votes. Therefore to be able to summarize the votes on a global level, the number of votes have to be translated, which is done as described followingly. The number of party votes per constituency are taken and divided with the respective number of seats. The resulting quotient gets rounded to the next number and indicates the amount of voters per list. The rounded quotients, can then be summed across all districts and divided by the cantonal divisor to get each parties total seats. The divisor itself is set by an official authority, in Zurich for example, its the Department of the Government Councillor which is responsible for the elections. The responsible authority, needs to assure that with the chosen divisor exactly as many seats can be allocated as there are available. The number of seats which result per party, are rounded again to the next number. (Pukelsheim & Schuchmacher, 2011, p.1584-1585 ; Kanton Zürich, 2003, § 10 and 103)
- **Lower Apportionment** This part of the apportionment process, assigns each party's cantonwide seats from the upper apportionment to its district lists while preserving each district's predetermined seat total. First, the party votes in each district are divided by a district divisor chosen so that the rounded results add up to the district's seats (Pukelsheim & Schuhmacher, 2011, p. 1585). These initial allocations may not match the parties' cantonwide entitlements: in the 2011 Zurich election, the SVP would have received 55 seats instead of the 54 assigned in the upper apportionment . To correct such differences, every district list of a party is divided by the same additional list group divisor. A divisor above one reduces the party's unrounded quotients and can remove seats while a divisor below one can add seats. (Pukelsheim & Schuhmacher, 2011, p. 1586) This effect can be illustrated with a hypothetical numerical example based on the 2011 seat mismatch for the SVP mentioned above. Now it could be that the SVP list in District Y would have a quotient after division by the district divisor of 2.51. With the standard rounding, the SVP would receive three seats in this district. But if now as a list group divisor 1.01 is taken, then this quotient falls to $2.51/1.01 \approx 2.49$, and the SVP would get allocated only two seats in this district. Through that the mismatch would be resolved, considering that in no other district, the allocated number of seats for the SVP would drop too. If this is the case, then instead of 1.01 a smaller divisor would had to be choosen until, only the amount of seats fall away, which are needed. Further might the change in District Y cause, that the district gets allocated one seat less then it should have. The party and district divisors are then adjusted in turn until every party receives its cantonwide entitlement and every district retains its fixed number of seats. (Pukelsheim & Schuhmacher, 2011, p. 1586)

To summarize, the upper apportionment in the Double Pukelsheim system reduces inequalities in electoral success caused by the high natural thresholds of small constituencies under the previous district-by-district Hagenbach-Bischoff system. It does so by pooling parties votes across the canton before seats are assigned to constituencies. This is also the main improvement of DP regarding the three dimensions of electoral equality in contrast to Hagenbach-Bischoff.

4.1.3.1 Trade-offs and Criticisms

Alike every proportional system, also the Double Pukelsheimer system has its flaws for which it gets criticized. One of the main weaknesses of the biproprtional apportionment compared to the HB System is its complexity. It gets mentioned throughout, that its relatively difficult to understand the system holistically and that there is profound knowledge needed, to be able to compute the seat allocation. (Bundeskanzlei, 2013, p. 19 ; Bundeskanzlei, 2025, p.24 ; Lüscher, 2024, p.20) Lüscher tough mentions, that the usage of list alliances under the priorly used Hagenbach-Bischoff method also led to a high complexity situation. This especially since, beforehand the election it was impossible to estimate, which party would profit from the alliance. (Lüscher, 2024, p.20) Another main point of criticism, are the so called "discordant seat allocations". The discordant seat allocations describe the phenomena when within a district a party gets allocated more seats than another party with more votes in that district. This phenomena can appear through the lower apportionment process and the dual role of the system to achieve proportionality both on cantonal and district-level. The emerging situation, can cause confusement and frustration especially within the party which received more votes in that specific constituency. (Pukelsheim & Schuhmacher, 2004, p. 519) Not only for the parties themself it can be irritating, but also for the voters. This especially since in Switzerland elections often contain a personal vote component. If then a popular figure from one party, loses its seat to a less known figure from another party and district, this may raise concerns and lower the acceptance of thee system. A reaction to this flaw of the DP-System, was that the cantons of Zug, Schaffhausen and Graubünden implemented a majority clause, which ensures that the most popular party in the corresponding constituency receives atleast one seat. (Lüscher, 2024, p. 21) Friedrich Pukelsheim was already aware of this issue in 2004 and pointed out that the so called majority requirement can in rare ocassions be not fulfilled. On base of this he mentioned the possibility of the implementation of such a majority clause. (Pukelsheim & Schuhmacher, 2004, p. 520) Another issue of the system, is that different turnout level and different proportions of residents who are eligible to vote in the consituencies can determine the allocated mandates of other constituencies. To illustrate this, Lüscher provides an example: There are three Districts with the same size and number of mandates, but while District 1 produces 50% of all votes, District 2 and 3 only produce each 25%. Through this District 1 determines 50% of the canton-wide party result, while only having 33.3% of the mandates. Subsequently this leads to under- and overrepresentation in the corresponding districts. (Lüscher, 2023, pp. 578-579) One other point which was often discussed to be a possible weakness of the Double Pukelsheim, is a possible Party System Fragmentation. For example both reports of the Swiss Federal Chancellery in 2013 and 2025 mention the possible party fragmentation in parliaments. As a possible measure to prevent fragmentation, the implentation of thresholds is mentioned. (Bundeskanzlei, 2013, p.18 ; Bundeskanzlei, 2025, p. 24) Despite this concerns, Sandro Lüscher came in his Thesis to the perception, that the general trend of a rising amount of parties in swiss cantonal parliaments, is unrelated to the implementation of the Double Pukelsheim. He then also mentioned, that the inexistency of an effect, might come from that several cantons have implemented the by the Federal Chancellerey mentioned thresholds. (Lüscher, 2024, p. 135) Therefore it can't be conclusively stated, whether the reform would have had an impact on party system fragmentation without the implentation of thresholds or not.

Taken together, these criticisms illustrate the central trade-offs of Double Pukelsheim. Although the system improves canton-wide proportionality and reduces the number of

ineffective votes, it does so at the cost of greater computational complexity and potentially counterintuitive outcomes at the constituency level. Majority clauses and electoral thresholds can mitigate discordant seat allocations and party-system fragmentation, but may simultaneously weaken proportionality. The system therefore does not eliminate electoral distortions entirely, it rather prioritizes party proportionality and equal vote success over simplicity and fully autonomous constituency-level representation.

4.1.4 The Adoption of Double Pukelsheim in Swiss Cantons

In the following paragraphs the arguments brought up in each reform canton will be discussed in chronological order. The year in the beginning of each entry, marks the year of first application of the double Pukelsheimer in the corresponding canton.

- **2007:** The canton of Zurich as the initiatior of the development of the Double Pukelsheimer Algorithm, became the first canton to make use of it in 2007. The reform followed a 2002 voting-rights complaint about elections to Zurich's city parliament. The Swiss Federal Supreme Court partially upheld the complaint. It found that the thresholds in some districts with up to 33.3% were massively above the 10% threshold, which is seen as compatible with the prinicples of proportional representation. Further was it claimed, that thresholds vary to much between districts. The ruling required the city of Zurich to reform its electoral system, subsequently the canton of Zurich, which was facing the same issues as the city, implemented the Pukelsheimer Proporz for cantonal elections with a 5% district-level threshold for parties to enter parliament (BGE 129 I 185 E. 7.1.2, E. 7.6.2–7.6.3 and E. 8.3.2; Lüscher, 2024, pp. 9–10, 33-36 ; Pukelsheim & Schuhmacher, 2004, p.505). In 2021, Zurich added a 3% cantonwide threshold, first applied in the 2023 election. A change to how that threshold is calculated was adopted in 2025 and will first apply in the 2027 election (Regierungsrat des Kantons Zürich, 2024, pp. 2, 5; Regierungsrat des Kantons Zürich, 2025, p. 1).
- **2008:** Schaffhausen: In a 2004 popular vote, Schaffhausen's electorate approved a reduction of the cantonal parliament from 80 to 60 seats. Shortly after a motion calling to reform the electoral law to better protect smaller parties was submitted. After this the Schaffhausen Government proposed a reform of the electoral system and the implementation of the biproportional apportionment. The biproportional apportionment was also preferred, because of the Supreme Courts decisions in Zurich and Aargau. (Regierungsrat des Kantons Schaffhausen , 2007, pp. 1, 3, 13-14) Subsequently the parliament (2007) and the voters (2008) approved the changes. In 2015, the cantonal parliament added a majority clause, which ensures that the strongest party receives at least one seat in the corresponding district. (Lüscher , 2024 , pp. 42- 43) In addition to the majority clause, did the canton of Schaffhausen decide to prohibit list alliances (Kanton Schaffhausen, 1904/2019, Art. 2b para. 3).
- **2009:** Aargau's voters in the year 2003 approved a proposal in a popular vote, which wanted to lower the amount of parliamentary seats from 200 to 140. This decision naturally raised the thresholds in the districts and therefore a second constitutional amendment allowing to merge small districts was proposed and approved by the voters. However, the parliament decided against making use of it. As a consequence of this decision, the electoral equity wasn't given anymore. The Government of Aargau had to act and proposed to the Grand Council to either merge districts or to change to the Double-Pukelsheim system. The Grand Council though, again decided not to change anything. Shortly after this parliamentary decision the FSC issued in a verdict, that there are existing inequalities, arising from the

varying thresholds and district sizes. The court ordered that this needs to be changed. In September 2007, the Grand Council with a narrow majority then decided to implement the Double-Pukelsheimer. In 2011 a 5% local (district) and a 3% global quorum was added after approval of the Grand Council and the voters (Lüscher, 2024, pp. 38-40)

- **2014:** Nidwalden: In the year 2010, members of the Green Party in Nidwalden filed a complaint regarding the seat allocation process for the cantonal elections (Lüscher, 2024, p. 44). That complaint got rejected initially by the Cantonal Court but later the cantonal decision got overruled by the FSC. The Supreme Court rejected the argument that historical practice justified the allocation method (BGE 136 I 352 E. 4.; Lüscher, 2024, pp. 44–45). As a reaction to this decision the parliament of Nidwalden came up with multiple possible reforms. The most popular of the suggested reforms, was the change towards the biproportional system. The parliament and the electorate subsequently approved the implementation of the Double-Pukelsheimer. (Lüscher, 2024, p.45)
- **2014:** Zug: In 2010, representatives of several parties formed a complaint at the FSC regarding the differently sized districts. The Federal Supreme Court agreed on the appeals main point (BGE 136 I 376 E. 4.5). As a reaction to this decision the cantonal Government presented three different possibilities on how to adapt to the FSC ruling. Of which the responsible comission favoured the adoption of the Double Pukelsheim. Subsequently, the parliament agreed on the adaption of the Double Pukelsheim and added based on the model of Aargau a 5% district-level and 3% cantonal-level threshold. (Lüscher, 2024, p. 47-51) A competing proposal to retain the existing system was struck down by the FSC in 2013 (BGE 139 I 195 E. 4). In addition to the thresholds, the canton also implented a majority clause, ensuring that the most popular list within a district always receives at least one seat (Kanton Zug, 2006/2026, § 52e paras. 1a–1b). Voters approved the biproportional reform in September 2013 (Lüscher, 2024, p.51).
- **2016:** Schwyz: The reform process began with the approval of the new cantonal constitution by the electorate. In that it was foreseen, to keep the current electoral system, which means that each municipality is its own electoral district. (Lüscher, 2024, p. 53) In September 2011, the Cantonal Government prepared the Cantonal Council Election for 2012, against the decisions published there three individual complaints were raised at the FSC. The main complaint was that the thresholds in some constituencies were too high. The Court agreed on the complaints of the three parties partly and stated that the current electoral system doesn't meet the requirements of a proportional representation system (Urteil des Bundesgerichts 1C_407/2011 etc. vom 19. März 2012, Sachverhalt A–B und E. 5.3, 6). Later, the Federal Assembly, which is responsible for guaranteeing cantonal constitutions, also refused to grant its guarantee to the contested provisions of Schwyz's new constitution (Schweizerische Eidgenossenschaft, 1999, Art. 172 ; Lüscher, 2024, p.53). As a reaction towards this decision, two independent initiatives to resolve the issues were formed by political parties in the canton. One of them wanted the implementation of a majority system and the other argued for fairer proportional representation with the willingness to remove guaranteed seats for municipialities. The government reacted with the proposal to inaugurate the DP-System while keeping municipialites as districts and preserving their guaranteed seats. The parliament supported this counterproposal and decided to implement a 1% cantonal-level threshold. The electorate could then decide on either the majority vote initative or

the counterproposal. The counterproposal and with that the implementation of the DP was approved. (Lüscher, 2024, pp. 53-55)

- **2017:** Valais: In the year 2004, a complaint at the FSC was filed regarding the varying thresholds in the electoral districts. By that time, the FSC rejected the complaint and argued that there are historical reasons for the shapes of the electoral districts. (BGE 131 I 85 E. 2.3–2.5) Six years later an alliance from the Upper Valais proposed a popular initiative named "Every Vote Counts" and suggested with that the implementation of the DP with a total of only three districts. In 2011 the canton then set up a commission to develop proposals in these areas. (Lüscher, 2024, pp. 58–59) In 2012 multiple voters again filed a complaint at the Supreme Court. The FSC then changed its standpoint and ruled that the existing system violated the Federal Constitution (BGE 140 I 107 E. 4–5). Subsequently, the parliament approved a constitutional reform introducing biproportional seat allocation, with the addition of a 35 seats guarantee for the german-speaking Upper Valais. Voters narrowly rejected the package aswell as the initiative "Every Vote Counts" in 2015. To ensure lawful elections in 2017, the parliament adopted a temporary decree in March 2016 without the seat guarantee. The resulting system retained 14 subdistricts, grouped them into six larger districts, and applied biproportional allocation separately within each larger district. A year later, the Parliament incorporated these rules into the cantonal electoral law and with that made these arrangement permanent. (Lüscher, 2024, pp. 61–63) The Canton Valais further decided to implement a 8 % threshold which a party has to reach in atleast one of the 14 subdistricts to be included in the upper-apportionment allocation across six upper-level districts (Kanton Wallis, 2004b, Art. 154 Abs. 1).
- **2020:** Uri used a mixed electoral system: municipalities with one or two seats elected their representatives by Majorz, whereas those with at least three seats used proportional representation. A complaint against the seat allocation for the 2016 election led the Federal Supreme Court to rule that the allocation method within the Proporz municipalities violated electoral equality. The Court did not reject the mixed system itself, which it considered justifiable because it preserved personality-oriented elections and an autonomously elected representative for very small municipalities (BGE 143 I 92 Sachverhalt, E. 5–5.4 and E. 6.5 ; Kanton Uri, 2019, pp. 6–7). The cantonal government subsequently proposed Double Pukelsheim for all eight municipalities which had priorly used Hagenbach-Bischoff. During the legislative debate, however, the Landrat raised the threshold for the usage of the proportional system from three to five seats, moving Attinghausen, Flüelen, Seedorf, and Silenen to Majorz and leaving only Altdorf, Bürglen, Erstfeld, and Schattdorf in the Double Pukelsheim allocation. The parliamentary majority reasoned that a majority system is better suited for the four municipalities' small populations and strong autonomy. It also pointed to the weak role or absence of local party organizations, occasional uncontested elections, and electoral statistics suggesting that voters chose personalities rather than party lists. The government opposed the extension and warned of a high risk of failure, if it would come to another challenge of the electoral system in front of the FSC. Uri's voters though, approved both the constitutional amendment extending Majorz and the separate Proporzgesetz reform in May 2019 (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67). As of October 2026, Uri's revised electoral system remains in force and hasn't been overturned by the FSC.
- **2022:** Graubünden: Before the Implementation of the biproportional apportionment the canton was one of the last to use a majority based electoral system. This even though, that

throughout the years there were multiple attempts to introduce a PR system. A main reason for these attempts to fail, was the fear that the protection of local minorities and the 20 single-seat-constituencies would be weakened. (Lüscher, 2024, p.69) In 2017 then multiple parties and individuals filed a complaint at the Federal Supreme Court after the cantonal government had released the seat allocations for the districts in the upcoming election. The court partially upheld the complaint and it especially stated that the seat allocation for the smallest district (Avers) and the six largest districts needed to be revised. Specifically the Court found that the guaranteed seats for Aver's 160 eligible voters violated equality of voting weight, while majoirtiy voting in the six largest constituencies unjustifiably impaired the equal effect of votes on seat allocation. (BGE 145 I 259 E. 7.3 and E. 8.5.3) As a response to the FSC decision, the Government Council launched a consultation process after which, it presented nine different models. While the government preferred the DP-System, the municipalities and large parties favoured a mixed electoral model. The willingness of the Socialdemocrats to withdraw a separate initative helped build support for Double Pukelsheim. The parliament then chose the Double Pukelsheim as a compromise, since it made the seat allocation more proportional while keeping representation for small districts. A 3% canton-level threshold was additionally introduced. Moreover added the canton a majority clause similar to the one in Schaffhausen and Zug (Kanton Graubünden, 2021/2026, Art. 28 paras. 2–4). In 2021 the electorate voted in favor of implementing the DP-System (Lüscher, 2024, pp. 71-72).

- **2027:** Basel-Landschaft: Unlike other cantons, in Basel-Landschaft there were no implicit quorums of more than 10%, which means that the law complied with the FSC ruling. In 2015 though the parliamentarian Klaus Kirchmayr requested the cantonal government to examine the comparative effects of the current system with the Double Pukelsheimer. The Government didn't see any need for a reform, the parliamentarian though kept working on the topic and demanded the establishment of a special comission to review the topic. Although the parliament also rejected this proposal, it instructed its executive board to work with external experts to examine how the electoral system could better reflect party vote shares while preserving elected representatives' local ties. A few years later in 2022, the executive board came to the conclusion, that it recommended the implementation of the DP. Through the support of the parliament, the government was then instructed to draft a revision of the electoral law. (Lüscher, 2024, p.86) In the year 2024 the government then finally submitted the bill and the Cantonal Council voted in favour of it (Regierungsrat des Kantons Basel-Landschaft, 2024 ; Landrat des Kantons Basel-Landschaft, 2024, p. 1052). The reform was subject to a mandatory referendum and the canton's electorate voted in favour of the implementation in 2025 (Kanton Basel-Landschaft, 2025b). The canton further decided to implement a 3% canton-level and a 5% district-level threshold and stipulated that each constituency receives one guaranteed seat (Kanton Basel-Landschaft, 1981/2025, Art. 39a and Art.49 lit.a). The first election under the new system will be held in 2027 (Kanton Basel-Landschaft, 2026).

To summarize, in 8 of 10 reform cantons, a raised complaint at the FSC led to a later reform. Only in the cantons of Schaffhausen and Basel-Landschaft did the reform process take place without a Federal Supreme Court ruling. Nevertheless the canton of Schaffhausen's decision was directly influenced by the FSC's decisions in Aargau and Zurich. Throughout all complaints, the concern was electoral equality. In seven of them excessively high natural

thresholds in proportional constituencies were central. While in Graubünden the concerns were mainly raised by the use of of the majority system for its large districts and the allocation of seats among constituencies. This demonstrates the influence of the 2002 Federal Supreme Court decision regarding Zurich's City Parliament.

Beyond their different paths to reform, the cantons also implemented Double Pukelsheim under substantially different institutional conditions. The number of seats included in the Double Pukelsheim allocation varies from 37 in Uri where the system applies only to four municipalities to 180 in Zurich (Standeskanzlei Uri, 2020, p. 1; Kanton Zürich, 2003, § 88). Moreover, some cantons introduced explicit local or canton-wide electoral quorums, whereas others did not, potentially limiting smaller parties' ability to benefit from the more proportional allocation method. These arrangements have also changed over time—for example through the later introduction of thresholds or majority clauses—while party competition and vote distributions vary between elections. This institutional variation motivates the exploratory comparisons below: whether smaller parties' seat-share gains differ by the number of seats allocated through Double Pukelsheim, by the presence of an explicit quorum, and across successive elections within the same canton.

4.2 Literature Review

Building on the institutional mechanisms described above, the existing empirical literature generally finds that Double Pukelsheim improves vote–seat proportionality and benefits parties that were disadvantaged by small electoral districts. However, the studies differ substantially in their research designs, geographical coverage, and treatment of counterfactual electoral rules.

Pukelsheim and Schuhmacher provide an early counterfactual comparison based on the 2011 Zurich election. They apply the former Hagenbach–Bischoff method to the observed votes while assuming four hypothetical list alliances based on political orientations and historical alliances. Under this counterfactual, the larger SVP, SP, and FDP receive additional seats, primarily at the expense of smaller parties such as the BDP and Alternative List. Their calculation also indicates considerably greater inequality in the electoral success of votes under Hagenbach–Bischoff. The analysis nevertheless concerns only one election, and its results depend partly on assumptions about which list alliances parties would have formed. (Pukelsheim & Schuhmacher, 2011, pp. 1590–1592)

Lüscher conducts a similar comparison for the 2016 Schwyz election. Under his Hagenbach–Bischoff counterfactual, which does not assume any list alliances, the dominant SVP and CVP would have benefited. Whereas the left alliance and especially the GLP benefited from the implementation of the Double Pukelsheim. Most notably, the GLP's three observed seats would have been reduced to zero under Hagenbach–Bischoff. (Lüscher, 2024, p. 13) This example illustrates the potentially substantial effect of the allocation rule on an individual smaller party, but it does not establish whether effects of this magnitude occur systematically across cantons and elections.

Leuzinger broadens the comparison by examining the most recent election in each of the seven cantons that had introduced Double Pukelsheim by 2018. His study is the closest precedent for the present analysis because it calculates hypothetical seat distributions under the former cantonal electoral systems. However, it covers only one election per canton and assumes that the list alliances observed in the final pre-reform election would have persisted (Leuzinger, 2018,

p. 42). The present thesis does not carry these historical alliances forward. Consequently, the two counterfactual specifications can produce different Hagenbach–Bischoff seat allocations. This illustrates the sensitivity of counterfactual results to assumptions about unobserved alliance formation. Leuzinger also mentions that he observed, that mainly smaller parties profited from the transition and that around 10% of the mandates mechanically change the party, through the transition (Leuzinger , 2018, p.42).

Taken together, the counterfactual calculations by Pukelsheim and Schuhmacher, Lüscher, and Leuzinger constitute the closest methodological precedents for the present thesis. They apply a former allocation rule to votes observed under Double Pukelsheim and compare the resulting counterfactual seat distribution with the actual allocation. Their comparisons therefore hold the observed post-reform vote distribution constant and isolate the difference produced by the allocation rules.

The interpretation of this difference can be clarified using Fiva and Folke's framework for separating the mechanical and psychological effects of electoral reform. Fiva and Folke examine a reform of Norwegian municipal elections, first applied in 2003, that replaced D'Hondt with modified Sainte-Laguë. (Fiva & Folke, 2016 , p.266) Their principal estimate of the mechanical reform effect compares the actual pre-reform allocation with a counterfactual application of the new formula to the same pre-reform votes. They also apply the former and new formulas to the same post-reform votes to examine how behavioral responses may alter the magnitude of the mechanical effect. (Fiva & Folke, 2016, pp. 270–272) The Swiss studies described above, and the present thesis, follow this latter computational approach: the former formula is applied counterfactually to votes observed under the new system. This measures the mechanical allocation difference conditional on the observed post-reform vote distribution, but it does not estimate how parties and voters would have behaved had the former system remained in place or identify the reform's complete causal effect.

Rather than recalculating individual elections, Lüscher's principal empirical analysis estimates how cantonal political outcomes would have developed without the reforms. He finds that Double Pukelsheim substantially reduced vote–seat disproportionality, with stronger improvements in cantons that previously had smaller average district magnitudes (Lüscher, 2024, pp. 129–143). However, Lüscher finds no significant increase in the effective number of parliamentary parties or electoral candidacies. Voter turnout initially declined slightly, but this effect diminished over subsequent elections. These findings distinguish the reform's immediate mechanical consequences from its more limited behavioural effects on parties and voters.(Lüscher, 2024, pp. 143–145)

The existing literature therefore provides strong evidence that Double Pukelsheim improves proportionality, as well as several examples in which smaller parties gain seats. What remains less systematically examined is the magnitude and consistency of the conditional mechanical allocation difference across successive elections. Existing counterfactual studies either focus on individual elections or cover only one election per reform canton, while Lüscher's analysis examines aggregate disproportionality and broader political outcomes rather than the election-specific seat gains of smaller parties.

This thesis addresses that narrower gap by applying a consistent Hagenbach–Bischoff counterfactual to 31 elections in eight transition cantons. Two elections from Graubünden are additionally presented as contextual comparisons. Because Graubünden previously used

majority voting rather than Hagenbach–Bischoff, these allocations are hypothetical and are excluded from the main transition estimate. The thesis holds observed post-reform votes constant, applies a common definition of smaller parties, and measures their change in seat share for every election. In contrast to Leuzinger (2018) and Pukelsheim & Schuchmacher (2011), it does not assume that hypothetical or historical pre-reform alliances would have persisted. The analysis therefore estimates the mechanical allocation difference conditional on the observed votes without introducing an additional behavioral assumption about counterfactual alliance formation.

4.3 Expected Effects of the Transition

Based on the theoretical considerations developed in the preceding sections, several effects can be expected from the transition of the Hagenbach–Bischoff method to the Double Pukelsheim system. Under a district-by-district Hagenbach–Bischoff allocation, small district magnitudes create high natural thresholds for winning a parliamentary seat. Votes for parties which fail to win a seat in a district, won't contribute elsewhere. This produces ineffective votes and therefore disproportionately disadvantages smaller parties. The biproportional apportionment instead through the upper apportionment pools votes across the whole relevant area, before it distributes each party's seats among the districts. This should increase the equality of electoral success and reduce vote–seat disproportionality. Therefore mechanically, and conditional on the same observed votes, smaller parties that surpass the applicable legal thresholds are expected to receive a larger seat share under Double Pukelsheim, while the seat advantage previously enjoyed by larger parties should decline (Pukelsheim & Schuhmacher, 2004, pp. 506–508, 513-514 , 518).

The magnitude of this mechanical difference should depend particularly on district magnitude under the Hagenbach–Bischoff comparison. The fewer seats allocated in a district, the larger the vote share a party generally needs to obtain mandates. Double Pukelsheim should therefore produce larger gains for smaller parties where the former or counterfactual Hagenbach–Bischoff allocation consists of many low-magnitude districts than where seats are already allocated in relatively large districts. This expectation is consistent with Lüscher's finding that the reduction in disproportionality was stronger in cantons with smaller pre-reform average district magnitudes (Lüscher, 2024, pp. 132–133).

The effect is also expected to depend on the specific institutional design adopted by each canton. Reflecting Switzerland's federal structure, the reformed cantons differ in their use of canton-wide and district-level legal thresholds, majority clauses, and other supplementary adjustments. Explicit electoral thresholds can restrict gains among smaller parties by excluding lists that do not reach the required vote share and can therefore partly offset the greater proportionality produced by the upper apportionment (Bundeskanzlei, 2013, p. 18; Lüscher, 2024, pp. 107, 135). The expected effect of majority clauses is less unambiguous because they protect the strongest list within a district rather than parties of a particular size, thus they may modify individual district allocations without systematically benefiting either smaller or larger parties (Pukelsheim & Schuhmacher, 2004, p. 520).

Valais and Uri are particularly important special cases because Double Pukelsheim does not cover the canton through a single canton-wide upper apportionment. Valais conducts separate upper apportionments in six electoral regions, so votes aren't pooled across the entire canton. This is expected to limit the proportionality-enhancing effect relative to a fully canton-wide

implementation, particularly for parties whose support is dispersed across regions (Lüscher, 2024, p. 57). In Uri, Double Pukelsheim applies only to the four biggest muncipialities, while the remaining ones use Majorz. This limits the territorial and numerical scope of the reform, although it does not necessarily reduce the mechanical effect among the four municipalities included in the upper apportionment (Kanton Uri, 2019, pp. 4, 10–11; Lüscher, 2024, pp. 64–67).

The total number of seats allocated through Double Pukelsheim may provide further context. All else equal, a larger number of seats allows vote shares to be translated into seat shares more finely (Lüscher, 2024, p. 106). However, parliament size does not generate an unambiguous expectation for the difference between the two allocation methods because the Double Pukelsheim and Hagenbach–Bischoff allocations being compared distribute the same total number of seats. How those seats are divided among districts under Hagenbach–Bischoff is the more direct mechanism. The differences between parliaments of various sizes, are therefore only presented as exploratory context.

Comparisons across successive elections also provide exploratory context by examining whether smaller parties' mechanical seat-share gains persist or vary over time. No directional trend is hypothesized, as vote distributions, the composition of the smaller-party group, and electoral rules can change between elections

The estimated difference may additionally depend on the electoral rules used to construct the counterfactual outcomes, including the treatment of list alliances and canton-specific allocation provisions.

Together, these expectations provide the foundation for the following hypothesis and exploratory questions concerning differences in smaller parties' seat shares.

5 Hypotheses

To answer the research question, the thesis tests one main hypothesis concerning how the transition from the Hagenbach–Bischoff system to Double Pukelsheim mechanically affects the seat shares of smaller parties. Complementary exploratory analyses evaluate whether this effect varies with district magnitude, the presence of an explicit electoral threshold, and successive elections since implementation within each canton. Together, these analyses identify both the immediate mechanical and potentially evolving consequences of the electoral-system reform.

5.1 Hypothesis 1

For each election e, smaller parties or electoral lists are those with less than 10% of the vote in the *analyzed area*. The variable *analyzed area* describes the whole canton except in Uri, where the *analyzed area* includes only the four municipalities using the DP-System. The 10% cutoff is chosen with reference to the Swiss Federal Supreme Court's benchmark that natural (implicit) thresholds should not exceed 10% when an electoral system is redesigned (BGE 131 I 74 E. 5.3–5.4; Bundeskanzlei, 2013, p. 8). This benchmark motivates the definition of smaller parties used in this analysis.

Let $S^{DP}e$ and $S^{HB}e$ denote the aggregated number of seats allocated to smaller parties in election e under Double Pukelsheim and the counterfactual Hagenbach–Bischoff system, respectively. Aswell as let M_e denote the total number of seats included in the comparison.

Smaller parties' combined gain in seat share, shown for each election in Figure 1, is calculated as follows:

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points .}$$

The hypothesis uses the unweighted mean, $\bar{\Delta} = \frac{1}{E}\sum_{e=1}^{E}\Delta_e$, so each election counts equally. The analysis also reports a mean weighted by the number of analyzed seats in each election for comparison.

This leads to the following directional hypothesis: $\boldsymbol{H_1}$**:** Across the analyzed elections, the mean mechanical difference ({}) is positive: electoral lists or list groups with less than 10% of the vote receive, on average, a larger share of the analyzed seats under Double Pukelsheim than under the counterfactual Hagenbach–Bischoff allocation.

5.2 Exploratory Questions

5.2.1 Exploratory question 1: Differences by district magnitude

The first exploratory question researches on the differences by district magnitude. Concretely it asks: Does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach–Bischoff allocation tend to be larger in cantons whose initial district structure was more fragmented? All district measures use a fixed historical baseline from each canton's last pre-reform election, with Schaffhausen adjusted to 60 seats. District size is measured with multiple parameters. The primary measure is the mean baseline district magnitude. Moreover is the median, the relative share of districts with less than 9 seats and the coefficient of variation between the districts reported. These additional measures add context and can reveal otherwise unobserved patterns for e.g. may the mean be influenced by a small number of unusually large or small districts. For canton c, with $D_{c,0}$ baseline districts and district magnitudes $m_{dc,0}$, the mean district magnitude is

$$\bar{m}_{c,0} = \frac{1}{D_{c,0}}\sum_{d=1}^{D_{c,0}} m_{dc,0} = \frac{M_{c,0}}{D_{c,0}}.$$

A lower baseline mean or median district magnitude is expected to be associated with a larger average gain for smaller parties. As a complementary measure, the analysis reports the baseline share of districts with fewer than nine seats,

$$q_{c,0} = \frac{1}{D_{c,0}}\sum_{d=1}^{D_{c,0}} \mathbb{1}(m_{dc,0} < 9).$$

The nine-seat cutoff follows the Swiss Federal Supreme Court's reasoning: under the natural-quorum approximation $1/(M+1)$, districts with fewer than nine seats impose a threshold above 10%. A larger baseline value of $q_{c,0}$ is therefore expected to be associated with a larger average gain (BGE 131 I 74 E. 5.3–5.4 ; Bundeskanzlei, 2013, p. 8). Because the number of transition cantons is small and their electoral rules differ, these relationships are interpreted descriptively. Graubünden is because of the priorly stated reasons only shown for context.

As a complementary measure, EQ1 examines inequality between district sizes using the coefficient of variation (CV). While the median captures the size of the typical district, CV

describes how widely district magnitudes differ relative to their mean. The CV is calculated followingly:

$$CV_{c,0} = \frac{\sigma_{c,0}}{\bar{m}_{c,0}},$$

where $\sigma_{c,0}$ is the population standard deviation of district magnitudes in canton c at the pre-reform baseline.

The motivation to include CV as a measure is based on the FSC reasoning in BGE 136 I 352 E. 3.5, where it identifies unequal district seat counts as problematic for the equality of electoral success. Equal to the other additional measures, this measure is shown to provide additional insight and not to test a specific hypothesis.

5.2.2 Exploratory question 2: Differences by DP seat count (Parliament size)

The second exploratory question examines the relationship between the parliament size and the average gain of smaller parties, with the following question: How is the number of seats allocated through Double Pukelsheim associated with the average gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach–Bischoff allocation?

For each canton, the election-level gains Δ_e are averaged first so that cantons with more observed elections receive no extra weight. The resulting canton-level mean is then examined in relation to the number of seats allocated through Double Pukelsheim. As a cutoff value for the two comparison groups 100 was chosen, since then the eight relevant cantons split equally. Because both allocation methods distribute the same number of seats within each election and district magnitude is the more direct mechanism, no directional relationship is hypothesized. The association is interpreted descriptively rather than as an isolated causal effect of the number of Double Pukelsheim seats. Graubünden is shown separately because its pre-reform system was Majorz and its Hagenbach–Bischoff allocation is hypothetical (Bundeskanzlei, 2013, Table 1).

5.2.3 Exploratory question 3: Explicit electoral quorums

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation vary between elections conducted with and without an explicit electoral quorum?

An explicit quorum is a legally specified minimum vote share that a party or list group must reach at at a district or cantonal level to qualify for seat allocation. Among the eight cantons of the main analysis, five have applied an explicit quorum. Whereas Aargau is a special case, since it implemented the quroum only from the second election under Double Pukelsheim on. The canton of Graubünden implemented an explicit quroum and is shown for context. Further details around the details of the implementation of explicit thresholds across the cantons are stated in Section 4.1.4. Unlike the natural threshold created by limited district magnitude, the explicit quroum directly excludes parties that do not satisfy the legal requirements.

Because such quorums may prevent smaller parties from benefiting from the pooling of votes under Double Pukelsheim, the following directional expectation is formulated:

On average, the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation is smaller in elections with an explicit electoral quorum than in elections without one.

This comparison is interpreted descriptively because the level and geographical application of the quorums differ across cantons and elections and coincide with other institutional differences.

5.2.4 Exploratory question 4: Change across successive elections

How does the gain in smaller parties' seat share under Double Pukelsheim relative to the counterfactual Hagenbach-Bischoff allocation evolve across successive elections within each canton?

This analysis extends Hypothesis 1 by following the seat shares under both allocation rules and their difference Δ_e from each canton's first election under Double Pukelsheim onward. It assesses whether the overall difference identified in Hypothesis 1 is persistent across elections or driven by particular election-specific outcomes.

Stable positive values of Δ_e would indicate a persistent mechanical advantage for smaller parties, whereas substantial fluctuations or changes in sign would suggest that the effect is strongly dependent on the circumstances of individual elections. The within-canton perspective further provides a clear separate image for each canton, without mixing their different electoral systems. Given that the smaller parties seat share, the party composition, aswell as the district magnitudes and electoral rules can change between elections, no directional trend can be hypothesized.

6 Research Design and Analysis

6.1 Research Design and Counterfactual Strategy

This thesis uses a deterministic counterfactual reallocation design. The main analysis covers all 31 cantonal parliamentary elections held under Double Pukelsheim in Switzerland through 2026 in the eight cantons in which Hagenbach–Bischoff previously was in use as the electoral system. The two elections from Graubünden are included as contextual comparisons since the canton had previously used a majority voting system. At the computational level, the observations are electoral lists within electoral constituencies and elections, while the primary unit of comparison is the election.

Each election is evaluated under two allocation regimes. The observed outcome is the official allocation under Double Pukelsheim, obtained from the relevant cantonal election authority. The counterfactual outcome is obtained by applying the specified former legal allocation rules to the same election's observed votes, electoral units, seat magnitudes, and reported list structure. Each election therefore serves as its own comparison.

The counterfactual is constructed in five steps.

First, the official election files are imported and standardized to obtain, for every election, electoral unit, and reported list or list group, the observed votes, seat magnitude, and official Double Pukelsheim seats. Party and list labels are harmonized across source files, and voter numbers or vote shares are reconstructed where not already directly published by the structure of the official data.

Second, the counterfactual inputs are defined by holding the observed votes, electoral units, seat magnitudes, and reported list configuration constant. Joint lists reported as one unit in the official results remain pooled while lists affiliated with the same party, but running independently remain separated.

Third, the specified former allocation regime is applied separately within each electoral unit. Most counterfactual allocations use the district-level Hagenbach–Bischoff method described above. Where the former cantonal rules contained additional rules such as, quoruoms, majority-clauses, or tie-breaking provisions, these are incorporated into the calculation. The applicable canton-specific rules and features are documented in Section 6.2.

Fourth, the counterfactual seats obtained in the individual constituencies are aggregated to the party-election level and merged with the official Double Pukelsheim seat totals.

Finally, the resulting datasets are validated by checking party coverage, vote and seat consistency, electoral-unit seat totals, and the total number of seats allocated under both regimes. Where the source data provide multiple measures of the official Double Pukelsheim result—such as upper-apportionment totals, lower-apportionment totals, reported list mandates, or elected-candidate counts—these measures are additionally reconciled. If exact ties between parties occur, they are if possible resolved with the applicable former cantonal laws. For the two cases in which the tie persisted, both possible seat allocations were presented for the affected parties. This assures, that a seat is never assigned unintentionally through e.g. the arbitrary ordering of rows. The two mentioned unresolved alternatives were observed in Graubünden 2022 and Schwyz 2024 and concern only parties above the 10% threshold and therefore do not affect the main smaller-party comparison.

The resulting contrast between the observed Double Pukelsheim allocation and the countrefactual is interpreted as a mechanical allocation difference conditional on the observed post-reform votes and not as the reforms complete causal effect. This since these votes were cast under Double Pukelsheim and the analysis cannot determine how voters, parties, number of candidates, campaign strategies, or list alliances would have differed if the former electoral regime had remained in force.

6.2 Counterfactual Assumptions and Scope

Applying the general counterfactual strategy requires several canton-specific decisions concerning electoral rules, list alliances, electoral units, and seat magnitudes. This subsection explains these decisions and defines the scope of the comparison.

Across the nine reform cantons, the counterfactual applies a district-level allocation to the electoral units retained from each observed election. In eight cantons, this generally reconstructs the formerly applicable Hagenbach–Bischoff regime, subject to the exceptions summarized in Table 1.[2] In Graubünden, which previously used Majorz, the Hagenbach–

[2] Aargau (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b); Staatskanzlei des Kantons Aargau, 2004, section "Neu: 140 statt 200 Mitglieder"); Graubünden (Kanton Graubünden, 2003/2018, Art. 27; hypothetical HB specification based on Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)); Nidwalden (BGE 136 I 352, E. 3.2); Schaffhausen (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37; Regierungsrat des Kantons Schaffhausen, 2007, pp. 2, 6); Schwyz (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, E. 5.2); Uri (Kanton Uri, 2018, Art. 22–23, "Bisher"; Kanton Uri, 2019, pp. 4, 11); Valais (BGE 140 I 107, Sachverhalt, pp. 108–109; Kanton Wallis, 2004a, Arts. 154–156); Zug (Kanton Zug, 2006, §§ 46–47, 55–56; BGE 136 I 376 E. 4.2, 4.4 and 5.1); and Zurich (Bentz, 2003, p. 2; Kanton Zürich, 1983/2005, §§ 87–88b)

Bischoff allocation is entirely hypothetical and is therefore excluded from the main transition estimate. In every case, the observed votes, electoral-unit seat magnitudes, and reported list structure are held constant. The common Hagenbach–Bischoff procedure is described in Section 4.1.2. This subsection therefore focuses on canton-specific adaptations and limitations of the reconstruction.

Importantly, the comparison in this thesis concerns the complete election-specific allocation regimes previously used in the cantons, rather than only the mathematical difference between the two apportionment algorithms. The observed outcome incorporates the Double-Pukelsheim rules applicable in the respective election, including any explicit electoral quorum or majority clause. Further the counterfactual applies the relevant former district-level rules to the observed votes and electoral structure. Consequently, differences in seat allocation may reflect both the change in apportionment method and associated differences between the applicable rule packages.

Table 1: Former electoral systems and counterfactual treatment by canton.

Canton	Pre-reform allocation	Counterfactual treatment	Canton-specific counterfactual rule
AG	HB in 11 electoral districts	Former HB allocation reconstructed	Standard HB procedure
GR	Majorz	Hypothetical district-level HB allocation in 39 electoral districts ; excluded from the main estimate	Hypothetical standard HB specification based on the basis Forumula in Aargau Law
NW	HB in 11 municipalities	Former HB allocation reconstructed	Standard HB procedure
SH	HB in 5 multi-seat districts, majority rule in 1 single-seat district	Former mixed district-level system reconstructed	Ceiling-based distribution number; relative majority in the single-seat district; statutory tie-breaking where required
SZ	HB in 30 municipalities	Former HB allocation reconstructed	Standard HB procedure; unresolved exact ties retained as alternatives
UR	HB in 8 municipalities, Majorz elsewhere	HB reconstructed for the 4 municipalities using DP	Tie-breaking by initial remainder and then party votes; unresolved alternatives retained
VS	HB in 14 districts	Former district-level HB regime reconstructed	8% district eligibility threshold: ineligible lists' votes remain in the distribution-number calculation
ZG	HB in 10 municipalities; Majorz in Neuheim	HB applied in all 11 municipalities; hypothetical for Neuheim	Standard HB procedure; hypothetical HB application in Neuheim
ZH	HB in 18 electoral districts	Former HB allocation reconstructed	Tie-breaking by party votes, followed by lot; unresolved lots retained as alternatives

Of all canton-specific provisions in Table 1 several do not affect the smaller-party outcomes. For example, were the statutory tie-breaking rules in Schaffhausen, Uri, and Zurich not triggered in any analyzed election. The Schaffhausen rule for calculating the distribution number though, produced differences in two cases: 5,022 rather than 5,023 in Klettgau in 2008 and 2,196 rather than 2,197 in Neuhausen in 2024. These differences changed neither the initial nor the final seat allocation. In Schwyz 2024 and in Graubünden 2022, an unresolved exact tie produced alternative HB allocations for parties above the 10% threshold. Thus it only affected individual party totals but not the aggregated seat total of smaller parties used in

the main analysis. The relative-majority allocation in Schaffhausen's single-seat district was nevertheless applied and remains part of the reconstructed fomrmer system.[3]

Beyond the statutory allocation rules, the counterfactual requires an assumption about unobserved list alliances. Unlike Leuzinger and Pukelsheim & Schuhmacher, this analysis does not carry historical or hypothetical alliances forward. List alliances are in this thesis seen as strategic and election-specific rather than fixed institutional parameters. This leads to, that election-specific, historical alliances cannot simply be assumed to persist across subsequent elections with different parties and vote distributions. In Zurich, for example, the last Kantonsrat election under Hagenbach–Bischoff took place in 2003. Assuming that the alliances formed in 2003 would still have applied in 2023 despite two decades of changes in the party system would therefore introduce a strong and unverifiable behavioural assumption. Some parties that formed list alliances in Zurich in 2003 no longer existed or contested the 2023 election. Conversely, parties such as the Green Liberal Party, which contested the 2023 election, did not yet exist in 2003. (Statistisches Amt des Kantons Zürich 2003, table 3; Statistisches Amt des Kantons Zürich 2026; Grünliberale Partei Kanton Zürich n.d.). Therefore the counterfactual preserves joint lists or list groups represented as a single unit in the observed election data. But it does not reconstruct historical alliances or assume new alliances between separately reported units. This provides a transparent counterfactual that holds the observed electoral configuration constant and estimates the mechanical allocation difference between the specified election-specific regimes.

The exclusion of unobserved list alliances is an important limitation of the counterfactual. Under Hagenbach–Bischoff, list alliances can pool votes during seat allocation and thereby improve the chances of representation for smaller lists (Pukelsheim & Schuhmacher, 2011, p. 1590). If the parties observed under Double Pukelsheim would have formed such alliances under the counterfactual HB regime, excluding them may understate smaller parties' counterfactual HB seat totals and consequently overstate their estimated gain under Double Pukelsheim. However, neither the composition of these hypothetical alliances nor the distribution of any resulting seat gains can be reliably determined. Alliance gains could accrue to larger partners or leave the aggregated representation of parties below the 10% threshold unchanged. The direction is therefore theoretically plausible but cannot be quantified uniformly across elections.

A second research-design choice concerns the electoral units and their seat magnitudes. In most cantons, the reform retained the former territorial districts or municipalities, meaning that the observed electoral units broadly correspond to those previously used under HB. The principal scope deviations are addressed in the special-cases callout below. Regarding the district magnitudes, the use of election-specific magnitudes is deliberate. District seat entitlements are generally based on population and periodically recalculated. Consequently, the magnitudes from the last pre-reform election would not necessarily have remained applicable in later elections even if Hagenbach–Bischoff had been retained. The counterfactual therefore holds each observed election's electoral units and seat magnitudes constant across the two regimes. It asks how the same votes would have been translated into seats under the former district-level rules, conditional on the contemporary distribution of seats among districts.

[3]These findings are based on the author's calculations of the official election data described in Appendix 1. Sources for the applicable legal provisions are provided in the preceding footnote.

The remaining deviations from this general approach are summarized below.

ℹ Treatment of the Special Cases

Graubünden. The HB allocation is hypothetical and applied using the common allocation procedure employed elsewhere in the analysis. It's basic formula is taken from the former Aargau Law (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). Thus the observations are shown in grey and excluded from the main transition estimate. In the 2022 election, appears an unresolved exact tie between the "FDP" and "SP_GRUENE" in Rheinwald, to resolve it an hypothetical lot would be needed. The tie, which doesn't affect the aggregated seat total of smaller parties, is therefore retained as an unresolved amlternative in the counterfactual.(Kanton Aargau, 2004, § 13; author's calculations using Kanton Graubünden, 2022) Robustness checks using Zurich's party-vote tie-breaker and Schaffhausen's ceiling-based quota preserved all possible allocations across the 39 districts in both 2022 and 2026 (Kanton Zürich, 1983/2005, § 88b; Regierungsrat des Kantons Schaffhausen, 1979, § 36).

Zug. Neuheim used Majorz before the reform. An exact Majorz counterfactual cannot be reconstructed because the results of a hypothetical second ballot are unobservable. Neuheim is therefore included using a hypothetical HB allocation.(BGE 136 I 376 E. 5.1; Kanton Zug, 2022a, 2022b, 2022c, 2026) The specific implementation approach adopted for Neuheim has no bearing on the main analysis.

Uri. Only Altdorf, Bürglen, Erstfeld, and Schattdorf used Double Pukelsheim in the elections analyzed here. They account for 37 of the Landrat's 64 seats (Standeskanzlei Uri, 2020, p. 1). Uri's vote shares, including the less-than-10% classification of smaller parties, are therefore based only on the four municipalities using Double Pukelsheim, not on the whole canton.

Valais. Unlike in the other cantons, upper apportionment is conducted separately within six electoral regions rather than canton-wide. Seats and votes are aggregated at the cantonal level only for this analysis; the actual allocation occurs independently within each region (Lüscher, 2024, p. 57).

Schaffhausen. The parliament was reduced from 80 to 60 seats when Double Pukelsheim was introduced. The counterfactual retains the 60-seat structure because comparing the two allocation methods requires the same number and distribution of seats (Regierungsrat des Kantons Schaffhausen, 2007, pp. 1-2, 6).

With the counterfactual rules and scope established, the next subsection describes the official election data and their standardization across cantons.

6.3 Data Sources and Processing

This subsection describes the official election data used in the analysis and explains how the heterogeneous cantonal source files were transformed into a harmonized dataset.

The official election results were obtained primarily from publicly available datasets published by cantonal state chancelleries and statistical offices. Where the data wasn't available online in structured formats, such as CSV or XLSX, they were requested from the responsible cantonal

authorities directly. All contacted authorities except the Canton of Schaffhausen provided data in either CSV or XLSX. Therrefore the Schaffhausen results had to be extracted from official PDF documents. The precise sources and elections covered are reported in Appendix Table 1. The granularity and organization of the source data vary considerably across cantons and elections, ranging from canton-wide files to separate district- or municipality-level results. In Schwyz, for example, the 2016 results were distributed across 30 municipality-specific excel workbooks, whereas the 2020 and 2024 results were each provided in a single workbook. Further did the structure of the files differ, some such as the Graubünden ones, were prepared for public or media use and through that readily accessible, while others had originally been designed for internal administrative calculations and where therefore less structured.

Party and list names were standardized using readable, clear and consistent abbreviations. The same political organization was assigned a common identifier across cantons and elections, some of the examples include "SVP", "GLP", "ALG", and "PIRATEN". Meaningful changes over time were nevertheless retained. In particular, "CVP" and "BDP" remain separately identified before their merger, while "DIEMITTE" is used only after the formation of the new party in 2021 (Schaub, 2022). In bilingual Valais, French- and German-language labels referring to the same party were harmonized, generally using the German abbreviation for consistency with the other cantons. However, "SP_GAUCHE_CITOYENNE" and "SP_GRUENE_UNABHAENGIGE" were retained as distinct identifiers because they represented separately reported lists with partly different organizations in the background.

The treatment of joint and affiliated lists follows the electoral units reported in the official results. Where several political organizations contested an election on a single joint list, that list was treated as one analytical unit. Examples include "SP_GRUENE" in Graubünden in 2022 and "FGA_AL_PDA" in Zurich in 2023. It's important to state that joint lists and list alliances used under Hagenbach-Bischoff differ fundamentally. When forming a joint list, parties run together as a single electoral list (party), while under list alliances the parties run separately and the votes of the alliance parties are pooled later for the seat-allocation. (Pukelsheim & Schuhmacher, 2011, p. 1590) In addition to joint lists, in some cantons also affiliated or special-interest lists ran as a independent list. This phenomena occurred especially strong in Schaffhausen were among others the following affiliated lists ran for the 2016 election : "SVP_KMU" (SVP's small and medium enterprises list),"SVP_SEN" (SVP's senior's list), "JUSO" (Young Socialists) and "JFSH" (Young Liberals) (Kanton Schaffhausen, n.d.). Since these lists entered the official Double Pukelsheim allocation separately, they were also retained as separate units in the analysis. The analysis therefore contains exactly the lists which ran in the corresponding election with the effect that this decision possibly affects the measured number and seat share of smaller parties.

Beyond the harmonization of party and list names, the cleaning process involved restructuring vote data, identifying the relevant electoral units, and assigning their election-specific seat totals. The source files were converted into comparable tables containing party votes and the seat magnitude of each electoral district. Where parties appeared in separate columns, as in Aargau, Graubünden, and Nidwalden, these columns were reshaped into the variables *parties* and *party_votes*. Candidate-level exports required a different treatment. In all four Zug and Schwyz's 2020 election, each candidate row repeated the corresponding list's vote and seat totals. The scripts first checked that these repeated values were consistent and then retained one observation per municipality and list, preventing the totals from being counted multiple

times. Source variables such as *list_votes* and *election_mandates* were thereby converted into *party_votes* and *district_seats*. The geographic preparation depended on the structure of the official files. In Zurich, municipal party votes were summed within the electoral districts identified by the source variable *Geschaeft*. Aargau's files already contained district totals, which were selected using the *DISTRICT* classification. Similarly, the Graubünden script extracted the existing district totals by carrying each *Wahlkreis* heading down to its corresponding *Total* row. Municipalities were retained as allocation units in Nidwalden, Schwyz, and Zug. Their names were extracted from the source tables or, for Schwyz in 2016, from the municipality-specific filenames. Uri's script explicitly selected the workbook blocks for Altdorf, Bürglen, Erstfeld, and Schattdorf. In Valais, the 14 districts were assigned to six electoral regions in R according to the official electoral structure, preserving the different geographic levels required for the district-level HB counterfactual and the regional DP upper apportionment.

District seat magnitudes were obtained from the available official information for each election. In Aargau, district names were matched to officially sourced seat vectors or separate seat-allocation tables provided by the canton officials. Valais used explicitly recorded district magnitudes for each election year. In Zurich and Graubünden, magnitudes were reconstructed by counting elected candidates within each district. In Uri and Schwyz's 2016 election, they were calculated by summing the official DP seats within each municipality.

The shared finalization function then produced one observation per standardized party or list code and election. It created an election identifier combining the canton abbreviation and election year, such as *ZH_23*, and joined the HB and DP seat totals by party. The final columns were *election_id*, *parties*, *vote_share*, *small_party*, *seats_hb*, *seats_dp*, *parliament_size*, and *seats_allocated_dp*. The last two variables distinguish the full parliament from the seats covered by the comparison, this because of the Canton Uri, where these values are 64 and 37, respectively. District identifiers, raw party votes, and district magnitudes served as intermediate inputs rather than columns in the final combined dataset.

Vote shares were derived using the information available in each canton's official election files. In Aargau, Zurich, Zug, and Valais, party votes were divided by the corresponding district's seat magnitude to obtain voter equivalents. These quotients were rounded to the nearest integer, with half values rounded upwards, before being summed by party across districts. Each party's vote share was then calculated as its aggregated voter number divided by the total voter number of all included parties, multiplied by 100. Nidwalden instead used the voter numbers supplied in the official upper-apportionment worksheet. Schaffhausen used the published voter numbers extracted from the official result PDFs. Further where there the reconstructed voter numbers compared with the published values using election-specific tolerances to accommodate differences in reporting precision. Other datasets supplied vote shares directly or required different denominators. In Schwyz, the 2016 shares were calculated from each party's total cantonal votes relative to the sum of votes for all retained party lists, excluding the category for ballots without a party designation. For 2020, the denominator was the total accounted votes, counted once per municipality and checked against the summed list votes. The 2024 Schwyz shares and the Graubünden shares were taken directly from the official cantonal summaries. Uri's published shares refer exclusively to the four relevant municpialites and could therefore be retained. The shared finalization function rounded all vote shares to four decimal places.

The Double Pukelsheim results were assembled separately within the canton-specific scripts from official seat tables, municipal allocations, or elected-candidate records. Valais required a separate reconstruction of the upper apportionment within its six electoral regions, followed by comparisons with published official results. Each script supplied its election-specific DP results to the shared finalization function, which joined them to the HB counterfactuals by standardized party identifier. Validation routines checked electoral-unit coverage and seat totals and, where alternative representations were available, reconciled candidate counts or aggregated lower allocations with reported list mandates or upper-apportionment totals. District-level consistency checks and final allocation totals were applied according to the available source structure. Uri's allocation checks covered the 37 proportional seats included in the analysis, while the full parliamentary size remained recorded as 64. Unresolved HB ties were retained as alternative allocations.

One of the consistency checks revealed an error in the Zurich 2011 election data. There the party-votes total for the Socialdemocrats, were dobuled across all constituencies. The mistakes was found through the comparison of the summed party votes with the reported unit totals and cross-checking candidate and supplementary votes. Followingly, the error was reported to the Statistical Office of the Canton of Zurich. They confirmed the mistake and resolved it, the underlying cause was that a duplicate SP entry in the list-to-party mapping table had caused double counting during the aggregation of the official municipal results. The corrected dataset was uploaded and is used in this analysis. (S.Graf, personal communication, 2026-09-11)

All processing was implemented in the canton-specific R scripts in *data_cleaning_cantons*, supported by the shared functions in *00_functions.R*. Each *process_canton_()** function combined its cleaned elections into a cantonal dataset. These nine datasets were subsequently appended row-wise using *bind_rows()* to create *all_elections*, preserving one observation per standardized party or list and election. Detailed procedures, source-specific exceptions, and validation arrangements are documented in Appendix 1, with the cleaning code provided in Appendix 2.

6.4 Measures and Analytical Strategy

The analyzed area comprises the whole canton except in Uri, where it covers only Altdorf, Bürglen, Erstfeld, and Schattdorf, the four municipalities using Double Pukelsheim. These municipalities account for 37 of the canton's 64 parliamentary seats. The variable *seats_allocated_dp* records the number of seats covered by the comparison, while *parliament_size* records the full size of the cantonal parliament. These values are equal in all analyzed cantons except Uri.

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with a vote share below 10% in the analyzed area. Youth lists and other affiliated lists are treated as separate analytical units whenever they enter the official Double Pukelsheim allocation separately. Accordingly, the term *party* refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. *Smaller parties* are identified separately for each election using their observed vote shares. The same set of smaller parties is then used when comparing the two seat allocations within that election.

The main analysis compares the combined seat shares of smaller parties within each election. For this purpose, the seats of all lists below the 10% vote-share threshold are summed separately under Double Pukelsheim and Hagenbach–Bischoff. Each total is divided by the

number of analyzed seats, recorded in *seats_allocated_dp*, and multiplied by 100 to obtain the corresponding seat share as a percentage. This produces one aggregate seat-share difference per election. Using *seats_allocated_dp* as the denominator ensures that the measure refers to the same area as the votes and seat allocations included in the comparison.

The election-level outcome is the difference between these two seat shares, denoted by Δ_e and defined in Hypothesis 1. Positive values indicate that smaller parties collectively receive a larger seat share under Double Pukelsheim while negative values indicate a smaller share. A value of five percentage points therefore means that their combined share of the analyzed seats is five percentage points higher under Double Pukelsheim. This also means, that a negative or positive aggregate difference, does not mean that each included party gains or loses seats respectively.

Hypothesis 1 is assessed using the unweighted mean of the election-level differences, giving each election equal weight. This summarizes the average difference across the analyzed elections. Consequently, cantons with more observed elections contribute more observations to the main mean. A supplementary mean weights each election-level difference by the number of analyzed seats. It expresses the total net seat gain of smaller parties across the included elections as a percentage of the total analyzed seats, giving greater weight to elections with more seats. The median and the first and third quartiles are reported alongside these means to describe the typical election-level difference and the spread of the middle half of the observations. The main summaries exclude Graubünden for the reasons set out in Section 6.2 its observations are presented separately for context.

For EQ1, election-level seat-share gains are first averaged within each canton. These canton-level outcomes are examined against mean district magnitude, median district magnitude, the share of districts below nine seats, and CV. District counts, mean and median district magnitudes, shares below nine seats, and CV are all calculated from the same fixed allocation in each canton's last pre-reform election, with Schaffhausen adjusted to 60 seats. These historical characteristics are held fixed when comparing the canton-level average gains across the analyzed elections. Uri covers the same four municipalities included in the HB–DP comparison, and Valais's district count refers to its 14 HB allocation districts. For each measure, cantons are grouped below or at or above its unweighted mean across the eight transition cantons. Group means give each canton equal weight, and the group difference is calculated as below-mean minus at-or-above-mean. Graubünden is shown only for context and excluded from cutoffs and group means.

CV divides the population standard deviation of the baseline district magnitudes by their baseline mean:

$$CV_{c,0} = \frac{\sqrt{\frac{1}{D_{c,0}} \sum_{d=1}^{D_{c,0}} (m_{dc,0} - \bar{m}_{c,0})^2}}{\bar{m}_{c,0}}.$$

Here, c denotes the canton and 0 its pre-reform baseline; $D_{c,0}$ is the number of districts, $m_{dc,0}$ their seat magnitudes, and $\bar{m}_{c,0}$ their mean. The denominator of the variance is $D_{c,0}$ because the analysis covers the full set of districts in the analyzed area. A higher CV indicates greater dispersion relative to average district size; a CV of 0.5 means that the standard deviation is

half the mean. All district measures are calculated once per canton from this baseline and are held fixed across its analyzed elections.

The historical reference year applies to all district measures. Historical seat allocations are documented in the cantonal publications and court judgments cited below.[4] Later district magnitudes may change, so the district measures describe associations with the initial structure. For Schaffhausen, the 2004 structure is adjusted to 60 seats using the published allocation in the government's reform report (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2). This is an adjusted historical baseline, rather than the actual 80-seat allocation of the 2004 election. The election-specific HB–DP allocations are unchanged.

For the comparison by Double Pukelsheim seat count, election-level differences are first averaged within each canton. Cantons are then grouped according to whether they allocate atleast 100 or fewer than 100 seats through Double Pukelsheim. The 100-seat cutoff serves as a descriptive grouping of the observed cantons. Group means are calculated from the canton-level averages, giving each canton equal weight irrespective of the number of observed elections. The difference between the group means is calculated as the mean for cantons with atleast 100 seats minus the mean for cantons with fewer than 100 seats. Graubünden is excluded from these group means. Given the small number of cantons and their differing electoral rules, the comparison is interpreted descriptively.

To examine successive elections, the seat-share difference Δ_e is plotted within each canton against the election number since its first election under Double Pukelsheim. Arrows identify the first election affected by the subsequent rule changes annotated in the figure. Because the smaller-party classification is recalculated for every election, this comparison follows the category of smaller parties, whose membership may change over time.

Still to add:

- Procedure for the quorum comparison, once the classification and weighting of observations have been settled.
- Explicit quorums: how to classify elections as having or lacking a quorum, and how to calculate and compare the group summaries.

6.5 Results

6.5.1 Hypothesis 1: Smaller parties' seat-share gain

Figure 1 compares smaller parties' combined seat shares under the observed Double Pukelsheim allocation and the counterfactual Hagenbach-Bischoff allocation for every cleaned election. Smaller parties are defined as parties with less than 10% of the vote share in the analyzed area. The x-axis reports their combined share of analyzed seats, while the y-axis reports their combined vote share. The grey line connects the two allocations derived from the same election, and each label reports the gain under Double Pukelsheim in percentage points. This seat-share difference is the outcome used to assess Hypothesis 1 and allows comparisons

[4]Sources by canton and reference year: Aargau 2005 (Kanton Aargau, 2004); Graubünden 2018 (Kanton Graubünden, n.d.); Nidwalden 2010 (BGE 136 I 352, E. 3.5); Schaffhausen 2004, adjusted to 60 seats (Regierungsrat des Kantons Schaffhausen, 2007, Table 1, p. 2); Schwyz 2012 (Urteil des Bundesgerichts 1C_407/2011 vom 19. März 2012, Sachverhalt A); Uri 2016 (Standeskanzlei Uri, 2016, p. 5); Valais 2013 (BGE 140 I 107, Sachverhalt A); Zug 2010 (Staatskanzlei des Kantons Zug, 2010, p. 92); Zurich 2003 (Regierungsrat des Kantons Zürich, 2002).

across elections with different numbers of analyzed seats. The corresponding comparison in absolute seat counts is provided in Figure 7 in Appendix 3.

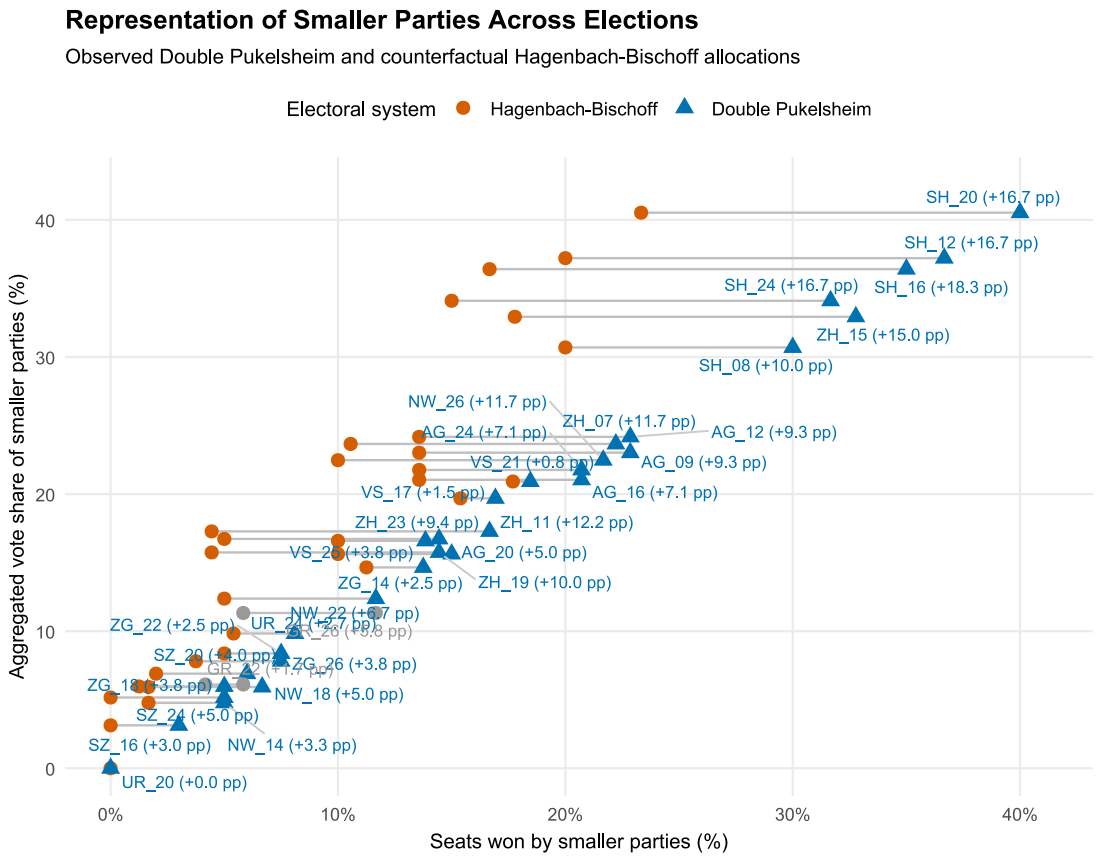


Figure 1: Percentage of analyzed seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff. Labels report the Double Pukelsheim gain in percentage points. Graubünden is shown for context only.

i Hypothesis 1 summary

Across all 31 elections (Graubünden excluded), the unweighted mean gain in smaller parties' share of analyzed seats was 7.6 percentage points (Q1: 3.5; median: 6.7; Q3: 10.8). The mean weighted by each election's number of analyzed seats was 7.8 percentage points. 16 individual elections exceeded five percentage points.

These results descriptively support Hypothesis 1 and indicate that smaller parties indeed in the 31 elections held receive a larger share of the analyzed seats under Double Pukelsheim than under Hagenbach-Bischoff. The effect is especially pointed out, through that in every election where small parties contested (every election but Uri 2020), they aggregately received a larger seat share. This finding is in accordance with Leuzinger's observation for the seven from him examined elections (Leuzinger, 2018, p.42). The weighted, unweighted mean and median are all relatively close to each other, indicating that the effect is not driven by a few elections with many seats and that the positive effect isn't determined by extreme outliers.

The average gain differs across the cantons for e.g. is there the Canton of Schaffhausen, with a stable high seat gain for smaller parties. There the high seat-share gain can be partially explained through the decision to treat affiliated parties as independent lists. Therefore these affiliated lists were classified as smaller parties, if they didn't reach the 10% threshold themselfs and with that higher the relative gain for smaller parties. Across the five Schaffhausen elections, the average effect of this decision is that the relative gain drops from 15.67 to 8.67 percentage points. Taking the effect into account, the mechanical transition effect in Schaffhausen is still relatively big compared to other cantons. Cantons with constantly relatively low stable seat gain shares on the other hand are for e.g. Uri, Zug and Schwyz.

To sum up the analysis, covers all 31 cantonal parliamentary elections held under Double Pukelsheim in Switzerland from the cantons that transitioned from Hagenbach-Bischoff. Across these elections, smaller parties received a higher aggregate seat share under Double Pukelsheim. This finding is based on the mechanical allocation difference conditional on the observed votes and specified counterfactual rules and therefore does not capture the whole transition effect. The next section examines whether the magnitude of this effect is associated with the cantons district magnitudes.

6.5.2 Exploratory question 1: Differences by district magnitude

This section examines how district magnitude varies across the cantons included in the analysis, considering both typical district size and variation within cantons. Table 2 summarizes these differences using district counts, the share of districts with fewer than nine seats, mean and median district magnitudes aswell as the coefficient of variation. All measures refer to the fixed historical baseline defined in Section 6.4 , with Schaffhausen adjusted to 60 seats. Graubünden is shown for context only.

Table 2: Pre-reform district structure by canton. Source: author's calculations using official cantonal allocations and court judgments.

Canton	Districts	Reference year	Mean seats	Pre-reform median	Below 9 seats (%)	Pre-reform CV
AG	11	2005	12.73	11.00	27.3	0.50
GR	39	2018	3.08	2.00	94.9	1.18
NW	11	2010	5.45	5.00	90.9	0.48
SH	6	2004	10.00	7.50	66.7	0.87
SZ	30	2012	3.33	2.00	90.0	0.89
UR	4	2016	9.25	8.00	50.0	0.38
VS	14	2013	9.29	8.00	50.0	0.59
ZG	11	2010	7.27	6.00	72.7	0.70
ZH	18	2003	10.00	10.50	38.9	0.39

Figure 2 relates each district measure to the canton-level mean gain in smaller parties' seat share. Colours distinguish cantons below and at or above the unweighted cross-canton mean of the respective measure. The cutoffs are defined by the mean of the eight relevant cantons and are the following: 8.42 seats for the unweighted mean magnitude, 7.25 seats for median

magnitude, 60.81% for the share below nine seats, and 0.60 for the CV. A CV of 0.6 means, that the standard deviation is 60% of the mean.

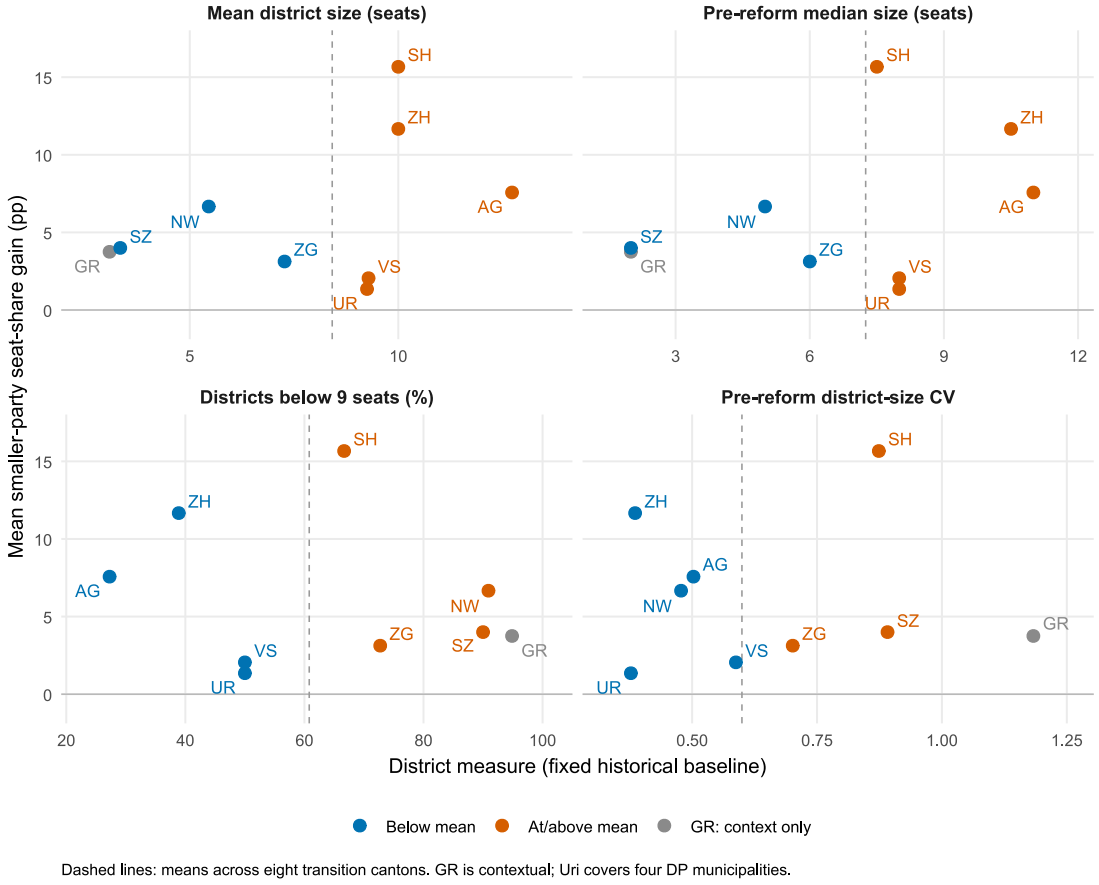


Figure 2: Mean smaller-party seat-share gain under Double Pukelsheim relative to counterfactual Hagenbach-Bischoff, by initial district magnitude and its dispersion. Dashed lines mark the cross-canton means and Graubünden is contextual.

Figure 2, shows no clear, consistent relationship between the former district structure and the relative smaller parties seat gain across the eight relevant cantons. For the mean and median district magnitudes, it was expected that cantons with smaller districts would see larger gains for smaller parties. Instead is the analysis slightly pointing into the opposite direction, as also visible in the later discussed Figure 3. Moreover is no clear pattern visible when comparing the groups above and below the 60.81% share of districts with a maximum of eight seats. The observation of a missing clear correlation, is further the case in the analysis on the pre-reform district-size cv. Followingly in Figure 3 , the same outcomes will be displayed as grouped dot plots with group means which makes the differences between the groups visually easier to assess. The missing paterns may be caused by the restricted amount of cantons and their different application of the Double Pukelsheim.

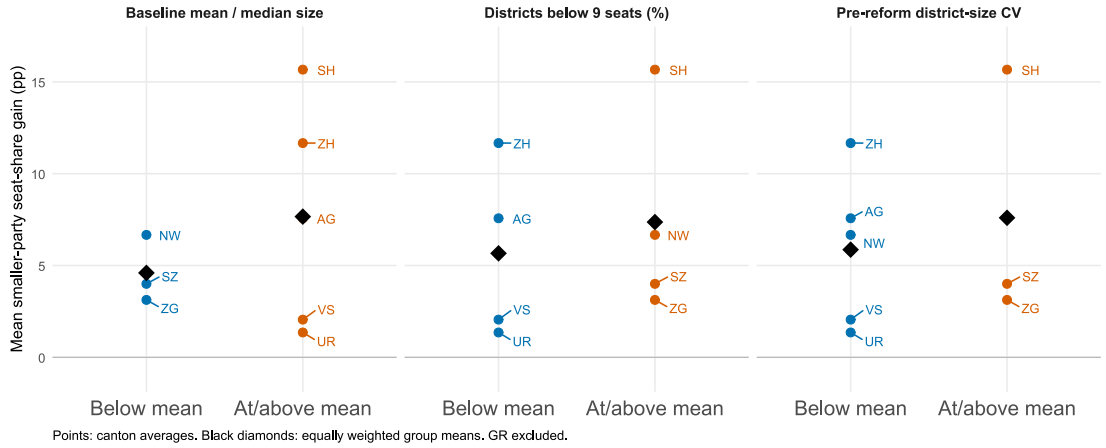


Figure 3: Canton-level mean smaller-party seat-share gains in groups below and at or above the cross-canton mean of each fixed historical district measure. Black diamonds denote equally weighted group means; Graubünden is excluded. The groups for the means and pre-reform median district magnitudes are equal and presented together.

For both mean district magnitude and pre-reform median district magnitude, the average gains for the below mean group are 4.60 and 7.66 percentage points for the at/above mean group. The at/above group results, are relatively spread out, while the below groups cantons are quite close to eachother. The below-minus-above difference is −3.06 percentage points and with that opposite to the expected direction.

For the share of districts below nine seats, the average gains for the below mean group are 5.66 and 7.36 percentage points for the at/above mean group. The higher-share group gains of 1.70 percentage points more are consistent with the expected direction. However is in the below group Schaffhausen included, whose comparatively high gain raises the group average. As discussed in Section 6.5.1 does the separate treatment of affiliated lists contribute to Schaffhausen's high gain. Therefore the result needs to be interpreted sensitively.

For pre-reform district-size variation, the average gains for the below mean group are 5.86 and 7.60 percentage points for the at/above mean group. The below-minus-above difference is therefore −1.74 percentage points. This means that the group of cantons with the higher variation in district sizes relative to their mean in average had higher gains. Here, too Schaffhausen's relatively high gain raises the group average, leading to that the result should be interpreted sensitively.

Taken together, the four district measures reveal no consistent pattern linking the initial district structure to smaller parties' gains. Although the comparison by the share of districts below nine seats follows the expected direction weakly, the mean- and median-magnitude comparisons point in the opposite direction. Individual canton gains also vary substantially within groups. Schaffhausen's comparatively high gain partly reflects the separate classification of affiliated lists as smaller parties, as discussed in Section 6.3 section. Also the higher average gain in the higher-CV group isn't enabling the establishment of a causal link. These descriptive comparisons therefore can not show any causal effect of district magnitude or its dispersion.

6.5.3 Exploratory question 2: Differences by DP seat count (Parliament size)

The following expoloratory question examines possible differences of the smaller party seat share gain between the different amounts of seats distributed with DP and with the parliament size. As priorly mentioned is the variable *seats_allocated_dp* used since in the canton of Uri only four municipialities use the Double Pukelsheim system and with that only 37 of the parliaments 64 seats are distributed by the new system. The comparison below averages elections within each canton before comparing the two groups defined by the number of seats allocated through Double Pukelsheim. Graubünden is as always only shown for context and excluded from the group means.

Table 3: Mean gain in smaller parties' share of analyzed seats by canton (percentage points). Source: author's aggregation of the official cantonal election datasets.

Canton	Group	Elections	DP seats (parliament)	Mean gain (pp)
ZH	At least 100 DP seats	5	180 (180)	11.7
AG	At least 100 DP seats	5	140 (140)	7.6
VS	At least 100 DP seats	3	130 (130)	2.1
GR	Context only (former Majorz)	2	120 (120)	3.8
SZ	At least 100 DP seats	3	100 (100)	4.0
ZG	Fewer than 100 DP seats	4	80 (80)	3.1
NW	Fewer than 100 DP seats	4	60 (60)	6.7
SH	Fewer than 100 DP seats	5	60 (60)	15.7
UR	Fewer than 100 DP seats	2	37 (64)	1.4

The mean of the canton-level gains is 6.3 percentage points in the group with atleast 100 Double Pukelsheim seats and 6.7 percentage points in the group with at most 100. The atleast-fewer difference is therefore −0.4 percentage points.

To visualize the analysis of a possible relationship between parliament size and the smaller parties seat-share gain, is visualized in a scatterplot. In Figure 4 each canton's average smaller-party-seat-share is plotted against the number of seats allocated through Double Pukelsheim. The two cantons are distinguished by colours, while Graubünden is displayed separately for context.

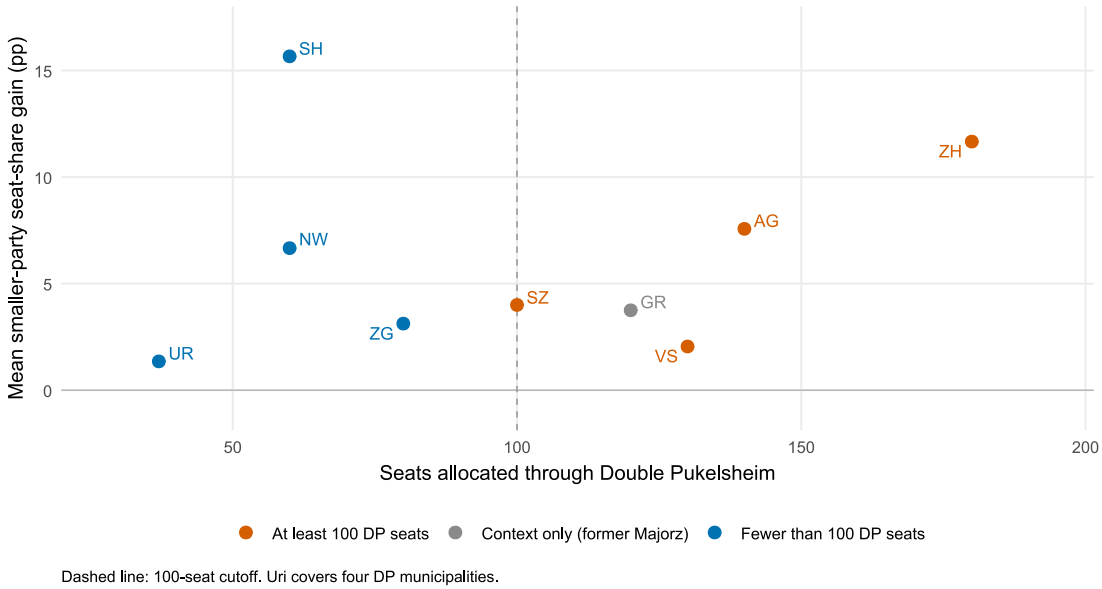


Figure 4: Canton-level mean smaller-party seat-share gain by the number of seats allocated through Double Pukelsheim. Graubünden is shown for context.

The Figure 4 reveals no clear consistent relationship between DP seat count and smaller parties average seat share gains across the eight cantons in the main analysis. Which is also illustrated through the fact, that the mean of both groups is with 6.3, respectively 6.7 percentage points close together. Noticeable again is the high seat-gain share of Schaffhausen, which was priorly discussed for e.g. in Section 6.5.1. Since this comparison is based on only 8 cantons operating under distinct electoral rules. Group averages may also conceal substantial variation between individual cantons, and the observed difference could reflect other institutional characteristics rather than the number of seats itself. The results therefore describe a difference within this sample but do not establish that a larger number of Double Pukelsheim seats causes greater gains.

6.5.4 Exploratory question 3: Explicit electoral quorums

Exploratory question 3 researches on whether the implementation of explicit electoral quroums have an influence on the smaller parties seat-share gain. The explicit quroums ensure that only parties which reach a specific vote-share are considered for the allocation of seats. This may prevent smaller parties from entering the parliament, which is why it's expected in this thesis, that in cantons with a quorum, the seat-share gain of smaller parties is reduced. This directional estimation is analyzed followingly:

Table 4: Situation regarding explicit electoral thresholds under Double Pukelsheim in the elections studied.

Canton	implemented Thresholds	relevant years
Aargau	none	2009
Aargau	5% district level or 3% cantonal level	2012, 2016, 2020, 2024
Graubünden	3% cantonal level	2022, 2026
Nidwalden	none	2014, 2018, 2022, 2026
Schaffhausen	none	2008, 2012, 2016, 2020, 2024
Schwyz	1% cantonal level	2016, 2020, 2024
Uri	none	2020, 2024
Valais	8% district level	2017, 2021 ,2025
Zug	5% district level or 3% cantonal level	2014, 2018, 2022, 2026
Zurich	5% district level	2007, 2011, 2015, 2019
Zurich	5% district level or 3% cantonal level	2023

Source: Author's compilation. Threshold rules are based on the sources discussed in Section 4.1.4. Election years refer to the elections included in the empirical analysis, whose data sources are documented in Section 10.1.

As documented in the table, out of eight analysis cantons, five applied a threshold. With Aargau being a special case, since the canton applied it only after their first election under Double Pukelsheim. The most often used explicit qourum is the combination of a 5% district-level and a 3% cantonal level threshold of which atleast one has to be reached. This combination was used in 9 of 19 elections held with an explicit threshold.

The difference is visualized followingly, in a bivariate analyisis presented as a grouped dot plot.

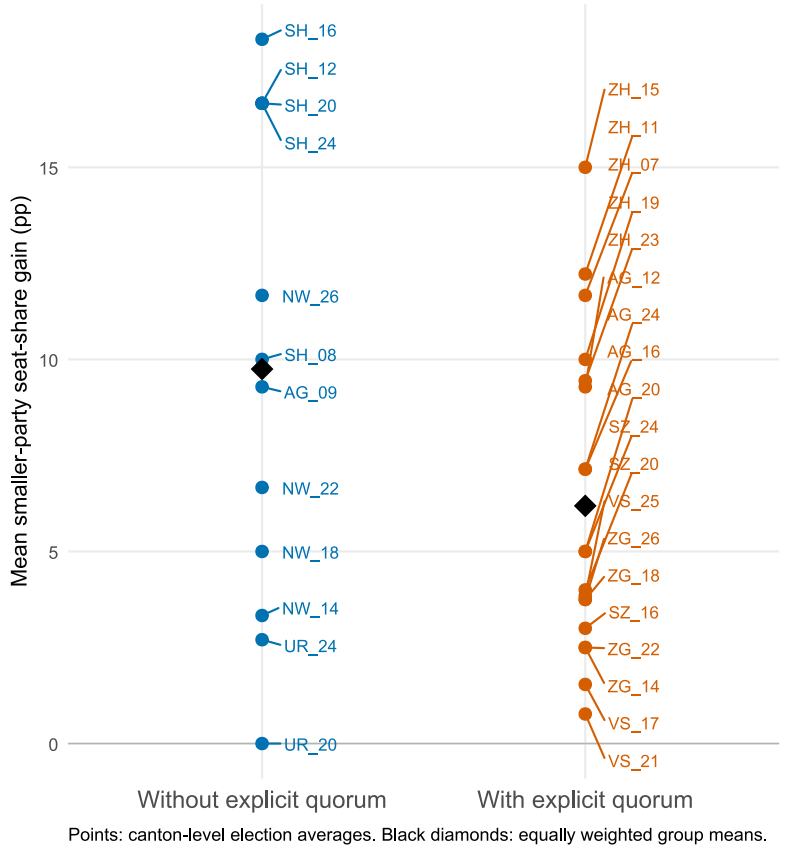


Figure 5: Smaller-party seat-share gains by explicit electoral quorum.

Figure 5 provides a descriptive comparison of smaller-party seat-share gains by the presence of an explicit electoral quorum. Average gains are higher in the group without an explicit quorum (9.7 percentage points) than in the group with one (6.2 percentage points). Both groups exhibit substantial variation. From which the canton of Schaffhausen sticks out, as mentioned in prior analysises.

Especially interesting, is Aargau, where in 2009 the election was held without a threshold and in the elections 2012, 2016, 2020 and 2024 with a threshold. The data shows that the seat-share gain for the pre-reform election 2009 and the first post-reform election 2012 is equally big with 9.3 percentage points. In the following elections, the seat-share gain dropped and was between 5.0 to 7.1 percentage points. The unchanged gain between 2009 and 2012 contrasts with the priorly seen difference between the groups. This suggests that the overall group difference does not consistently appear within individual cantons, although the two-election comparison cannot isolate the quorum's effect.

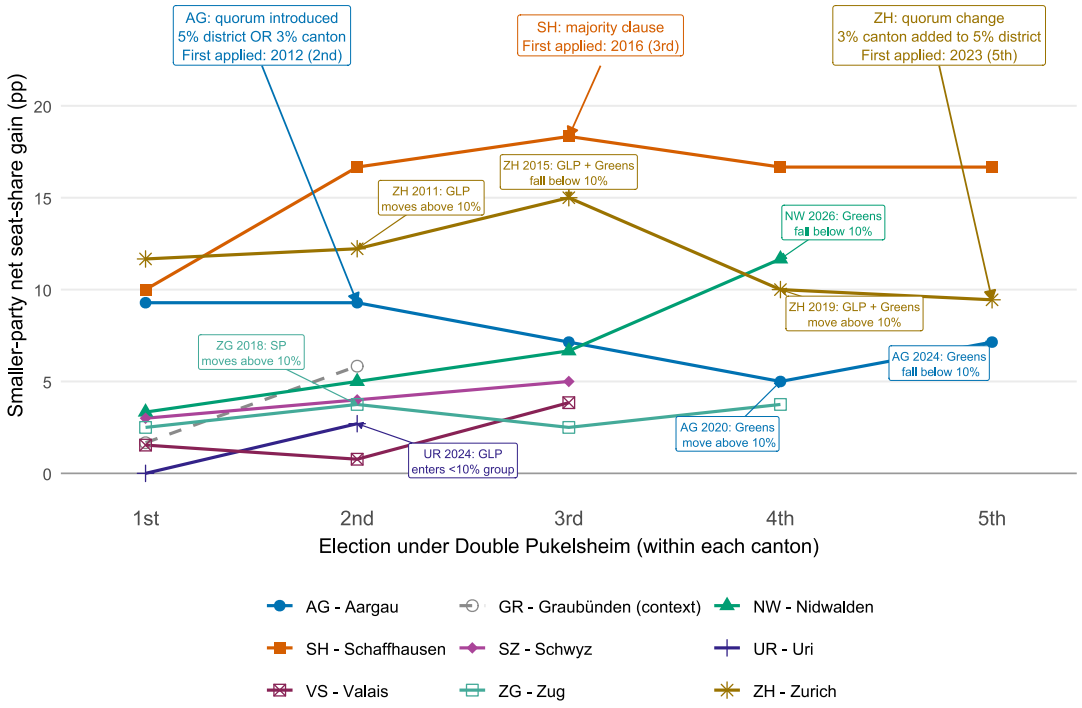
To summarize the analysis about the effect of the implementation of explicit quoroms, it can be stated that within the 31 relevant elections, indeed a difference can be seen. Especially does the direction of the seen pattern match the expectation, that explicit quorums limit smaller parties gains under Dobule Pukelsheim. However, because of the limited amount of elections held till date and the different application of thresholds and electoral law, the difference needs to be interpreted cautiosly.

6.5.5 Exploratory question 4: Change across successive elections

The preceding analysis showed that cantons differ in their implementation of the Double Pukelsheimer system. These institutional differences motivate a closer examination of within-canton patterns. The last exploratory question therefore examines how the smaller parties seat-share gain changed across successive elections within a canton. For that each canton and it's successive changes are analyzed separately. To possibly better understand changes, events such as electoral reforms and changes in the party composition are documented. Generally stable positive values of Δ_e indicate a persistent mechanical advantage for smaller parties, while substantial fluctuation or even changes in sign would suggest that the effect is strongly dependent on the individual circumstances of an election.

Since Figure 6 aligns every Double Pukelsheim election for each canton independently by their order, the first election under the system is numbered 1, the next is numbered 2, and so on. This means that the same election number represents different calendar years across cantons. As in the main hypothesis the outcome is the observed Double Pukelsheim seat share minus the counterfactual Hagenbach-Bischoff seat share, expressed in percentage points of the seats included in the comparison. The smaller-party category is recalculated for each election using the existing vote-share cutoff of less than 10%.

Regarding the events and their noteboxes, there are two different formats. The bigger boxes mark electoral rule changes, from which there were three relevant ones: Aargau's introduction of a 5% district-level or 3% cantonwide quorum in 2012 (second election); Schaffhausen's majority clause, applied in 2016 (third election); and Zurich's additional 3% cantonwide eligibility route, first applied in 2023 (fifth election). The Zurich rule supplements the existing 5% district-level quroum and as in Aargau, the parties have to only reach either 5% within a district or 3% on the cantonal level. These dates follow the institutional history in Section 4.1.4; Schaffhausen's amendment history records 1 March 2016 as the effective date of implementation (Kanton Schaffhausen, 1904/2019). The smaller boxes are marking the point, when there's a memberchange within the smaller parties group. Graubünden is shown as priorly only contextual.



Gain = observed DP minus counterfactual HB; smaller parties have less than 10% of the vote.
GR: hypothetical HB comparison. Uri: 37 proportional seats. Calendar years differ across cantons.

Figure 6: Smaller-party net seat-share gains across successive elections under Double Pukelsheim. Arrows identify the first election under each annotated rule change. Graubünden is contextual only. Sources: Large text boxes are based on the sources discussed in Section 4.1.4; small text boxes are based on the author's calculations.

i EQ4 summary

Excluding Graubünden, the average within-canton change in smaller parties' seat-share gain was +1.6 percentage points from election 1 to 2 (8 cantons); +1.0 percentage points from election 2 to 3 (7 cantons); −0.5 percentage points from election 3 to 4 (5 cantons); +0.5 percentage points from election 4 to 5 (3 cantons). Each average gives equal weight to the cantons observed in both elections of the respective transition.

Generally spoken, are the seat-share gains with some canton-specific fluctuations, relatively persistent. As seen in the analysis of Hypothesis 1, there is only one election with Uri 2020, in which there was no positive seat-share gain. This is the case, since there simply no a small party classified list ran. The strongest fluctuations can be seen for Schaffhausen, Zurich and Nidwalden. In Schaffhausen the seat-share gain rose from the first to the 2nd election from roughly 10 to roughly 17 percentage points, where it settled afterward quite stable. From the 2015 Zurich, to the 2019 Zurich election, the seat-share gain dropped by roughly 5 percentage points, a reason for this may be that both the GLP and the Green Party left the smaller-parties group, which means that there were less smaller parties for the 2019 election. In Nidwalden since the implementation, the share rose continously, from the 3rd to the 4th election, the share rose by 5 percentage points which might be influenced by that the Green Party entered the smaller parties group.

The figure describes within-canton variation in the mechanical allocation difference. Changes around an annotated election cannot by themselves establish an effect of the rule change: vote distributions, district magnitudes, and the composition of the smaller-party category may also have changed. Later election numbers contain fewer cantons, so comparisons across election numbers should follow the individual canton lines rather than interpret a changing pooled average.

7 Discussion

8 Conclusion

9 Bibliography

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10 Appendix

10.1 Appendix 1: Explanation Data Cleaning Process

The table below identifies the election data and legal sources used in the canton-specific cleaning scripts and counterfactual Hagenbach–Bischoff allocation. It specifies source titles, election coverage, file types, and the purpose of publications used to reconstruct or validate the results, together with the relevant legal provisions. Clickable citations and source titles lead to online sources where available; unpublished material supplied by email is identified explicitly. Author–year and case citations correspond to entries in the bibliography.

Canton	**Election data: description and source**	**HB allocation: legal basis and source**
Aargau	*Grossratswahlen: Detailresultate, ab 1985*: official district results for 2009, 2012, 2016, 2020, and 2024 (Statistisches Amt Kanton Aargau, n.d.). *Detailergebnisse der Grossratswahlen vom 8. März 2009* and *Jahresbericht mit Jahresrechnung 2012*: validation of cantonal seat totals (Department für Finanzen und Ressourcen des Kantons Aargau, 2009); (Regierungsrat des Kantons Aargau, 2013). *Gemeinsame Grossrats- und Regierungsratswahlen: Vorverfahren gestartet* (2012) and *Ausschreibung der Grossrats- und Regierungsratswahlen 2016*: district seat magnitudes (Staatskanzlei des Kantons Aargau, 2012; 2016).	*Grossratswahlgesetz (SAR 152.100), version in force from 1 November 2004*: initial allocation and allocation of remaining seats (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)).

Canton	Election data: description and source	HB allocation: legal basis and source
Graubünden	*Pressedaten Proporzwahl kantonal: Grossratswahlen*: official election-result Excel workbooks for 2022 and 2026 (Kanton Graubünden, 2022; 2026).	*Grossratswahlgesetz (SAR 152.100), version in force from 1 November 2004*: basis for the hypothetical HB allocation (Kanton Aargau, 1988/2004, § 12 and § 13(1)(a–b)). *Verfassung des Kantons Graubünden (BR 110.100), version of 1 January 2018*: confirms the former Majorz system (Kanton Graubünden, 2003/2018, Art. 27).
Nidwalden	*Wahlergebnis der Landratswahlen*: unpublished Excel workbooks for 2014, 2018, 2022, and 2026, supplied by the Kanzleisekretariat via email (Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026).	*BGE 136 I 352*: Federal Supreme Court description of the former district-level HB allocation (BGE 136 I 352, E. 3.2).
Schaffhausen	*Frühere Urnengänge (seit 2003)*: official result PDFs for 2008, 2012, and 2016 (Kanton Schaffhausen, n.d.). *Resultate der Kantonsratswahl*: official result PDFs for 2020 and 2024 (Staatskanzlei Kanton Schaffhausen, 2020; 2024).	*Proporzwahlverordnung vom 13. November 1979 (Amtsdruckschrift 2734)*: single-seat district rule, HB distribution number, and tie-breaking provisions. Original text supplied by the Staatsarchiv via email (Regierungsrat des Kantons Schaffhausen, 1979, §§ 2, 36–37).
Schwyz	*Kantonsratswahlen 2016: Gemeindeergebnisse*: 30 unpublished municipality-level CSV files supplied via email (Staatskanzlei Kanton Schwyz, 2016). *Kandidierendenresultate: Kantonsratswahlen 2020*: candidate-result CSV export (Staatskanzlei Kanton Schwyz, 2020). *Kantonsprotokoll der Erneuerungswahl Kantonsrat 2024*: election-result Excel workbook (Staatskanzlei Kanton Schwyz, 2024).	*Urteil des Bundesgerichts vom 19. März 2012 (1C_407/2011, 1C_445/2011 und 1C_447/2011)*: description of the former municipality-level HB allocation (Bundesgericht, 19. März 2012, E. 5.2).
Uri	*AuswertungKapitel2-2024*: unpublished Excel workbook containing the 2020 and 2024 results, supplied via email (Standeskanzlei Uri, 2024a). *Landratswahlen 2024: Statistische Auswertung*: official report used for party-label validation (Standeskanzlei Uri, 2024b).	*Synoptische Darstellung zum Gesetz über die Verhältniswahl des Landrats (Proporzgesetz)*: former allocation and tie-breaking rules in the column “Bisher” (Kanton Uri, 2018, Arts. 22–23).

Canton	Election data: description and source	HB allocation: legal basis and source
Valais	Official district-level Grand Council result workbooks for 2017, 2021, and 2025 (Excel files). *Grossräte 2017 – Resultate: Offizielle Endresultate*: validation of the 2017 results (Staat Wallis, 2017). Grand Council validation-commission reports: Bericht zur konstituierenden Session vom 19. April 2021 and Rapport relatif à la session constitutive du 14 avril 2025; validation of cantonal vote and seat totals. *Verteilung der Anzahl Grossratssitze pro Bezirk* (legislatures 2017–2021 and 2021–2025) and *Anhang zur Verteilung der Anzahl Grossratssitze pro Bezirk, Legislaturperiode 2025–2029*: district seat magnitudes (Kanton Wallis, 2016; 2020; 2024).	*Gesetz über die politischen Rechte (kGPR; SGS 160.1), version of 9 April 2017*: statutory eligibility threshold and district-level HB allocation (Kanton Wallis, 2004a, Arts. 154–156).
Zug	*Kantonsratswahl Kanton Zug* (2014), *Kantonsratswahl 2018*, *Kantonsratswahl 2022*, and *Erneuerungswahl für den Kantonsrat 2026*: official open-data candidate-result CSV exports (Kanton Zug, 2022b; 2022a; 2022c; 2026).	*Wahl- und Abstimmungsgesetz (WAG), original text of 28 September 2006*: former HB and Majorz provisions (Kanton Zug, 2006, §§ 46–47, 55–56). *BGE 136 I 376*: confirms Neuheim's former Majorz system; Neuheim is included using a hypothetical HB allocation (BGE 136 I 376, E. 5.1).
Zürich	*Archiv Wahlresultate Kantonsratswahlen*: official party-result and candidate-result Excel workbooks for 2007, 2011, 2015, 2019, and 2023 (Amt für Statistik und Daten Kanton Zürich, 2026).	*Gesetz über die Wahlen und Abstimmungen (Wahlgesetz; LS 161), version of 1 January 2005*: former HB allocation and tie-breaking provisions (Kanton Zürich, 1983/2005, §§ 87–88b).

Table 1: Election data sources and legal bases for the counterfactual Hagenbach–Bischoff allocation.

10.2 Appendix 2: Code for Data Cleaning

10.3 Appendix 3: Smaller parties' absolute seat counts

```
#show <fig-small-parties-absolute>: set figure(placement: none)
```

Figure 7 supplements the seat-share comparison used to assess Hypothesis 1 in Figure 1. It shows the absolute number of seats won collectively by parties below the 10% vote-share threshold under the observed Double Pukelsheim allocation and the counterfactual Hagenbach-Bischoff allocation. The grey line connects the two outcomes for each election. Because the number of analyzed seats differs across elections, these absolute counts provide context for the magnitude of the gains, while Hypothesis 1 is assessed using the difference in seat shares.

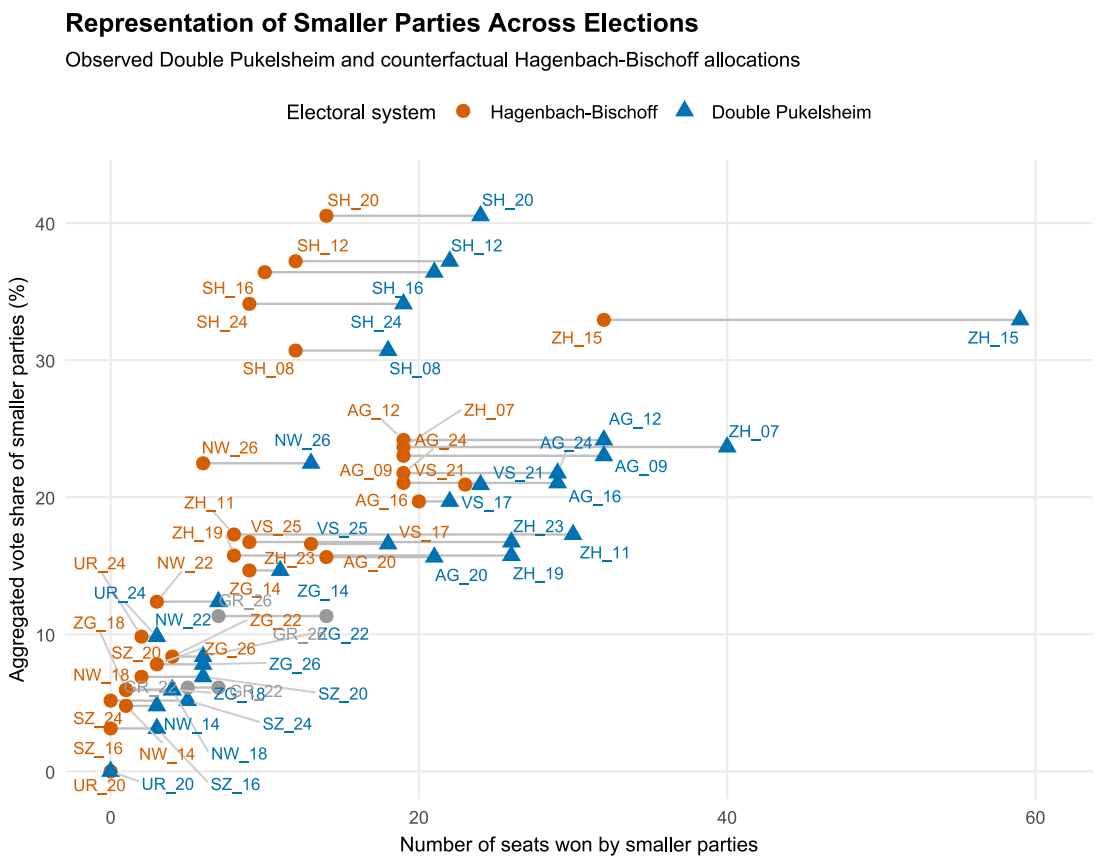


Figure 7: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties have less than 10% of the vote share in the analyzed area. Graubünden is shown for context only.

10.4 Declaration of AI-Assistance